

BFS Traversal

Step 1: Start at E

Legend

Complete

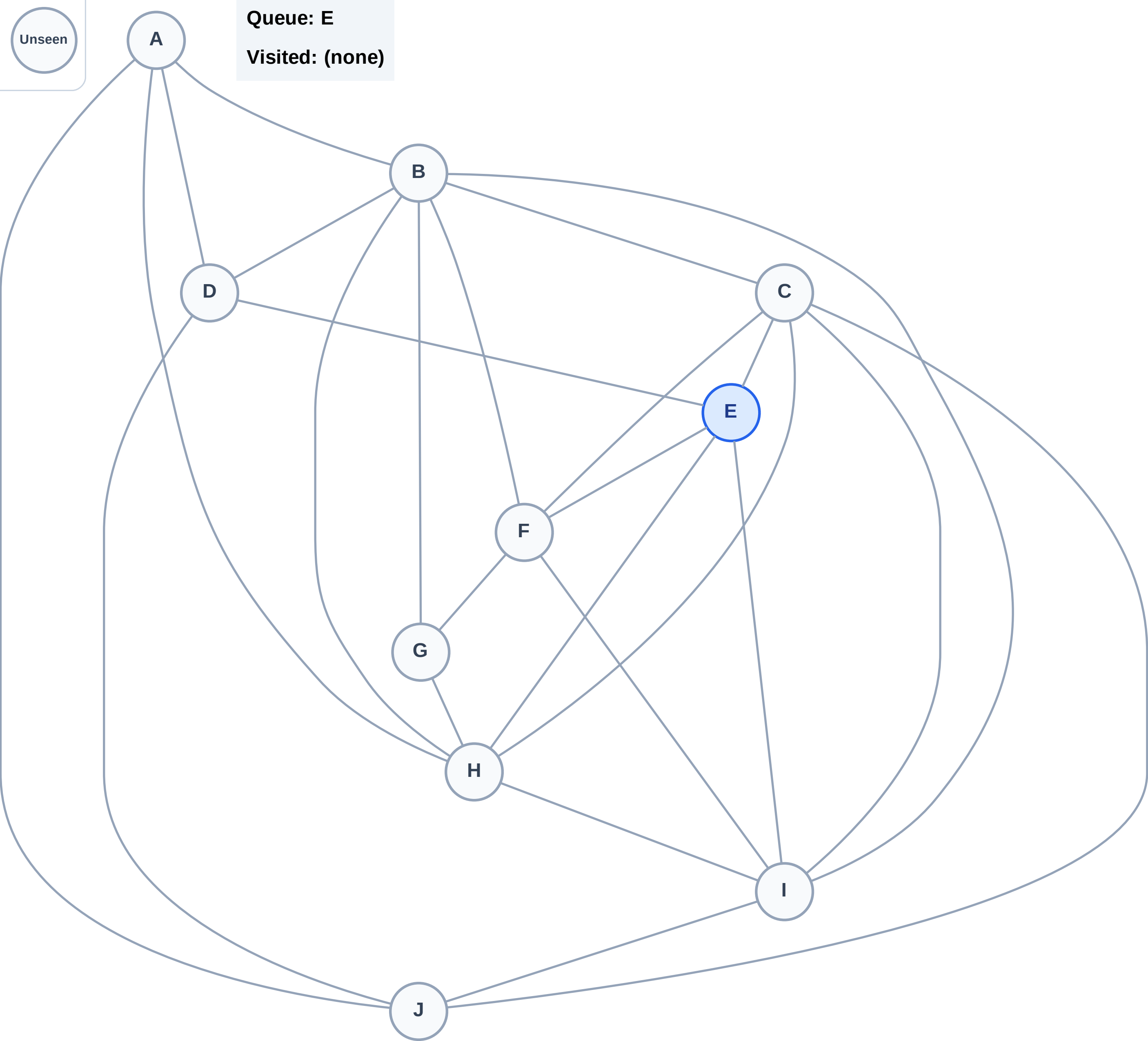
Processing

Discovered

Unseen

Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

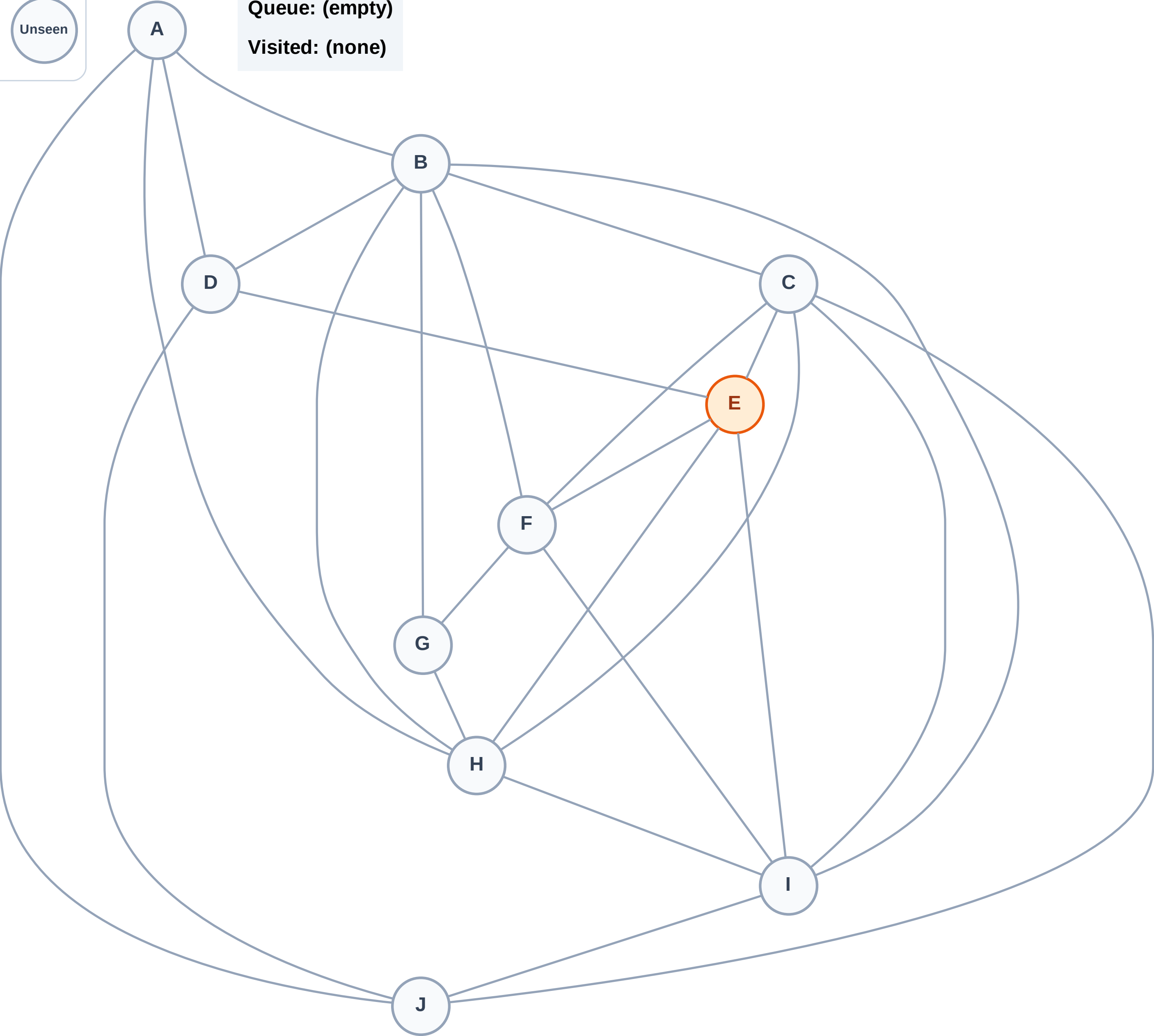
Complete

Processing

Discovered

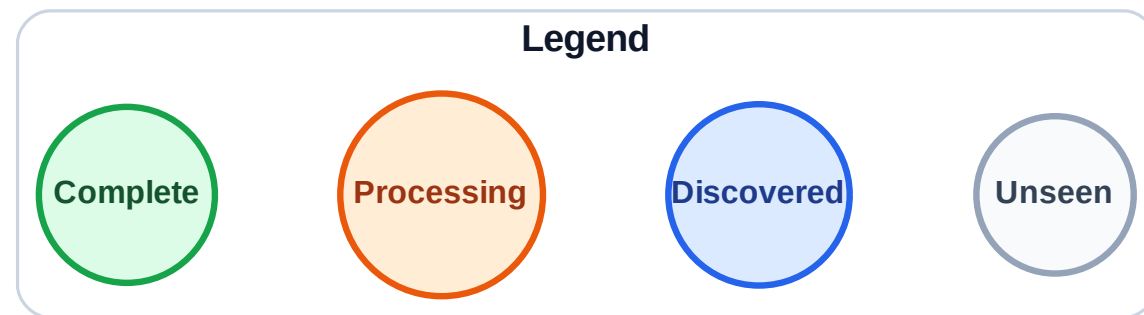
Unseen

Queue: (empty)
Visited: (none)

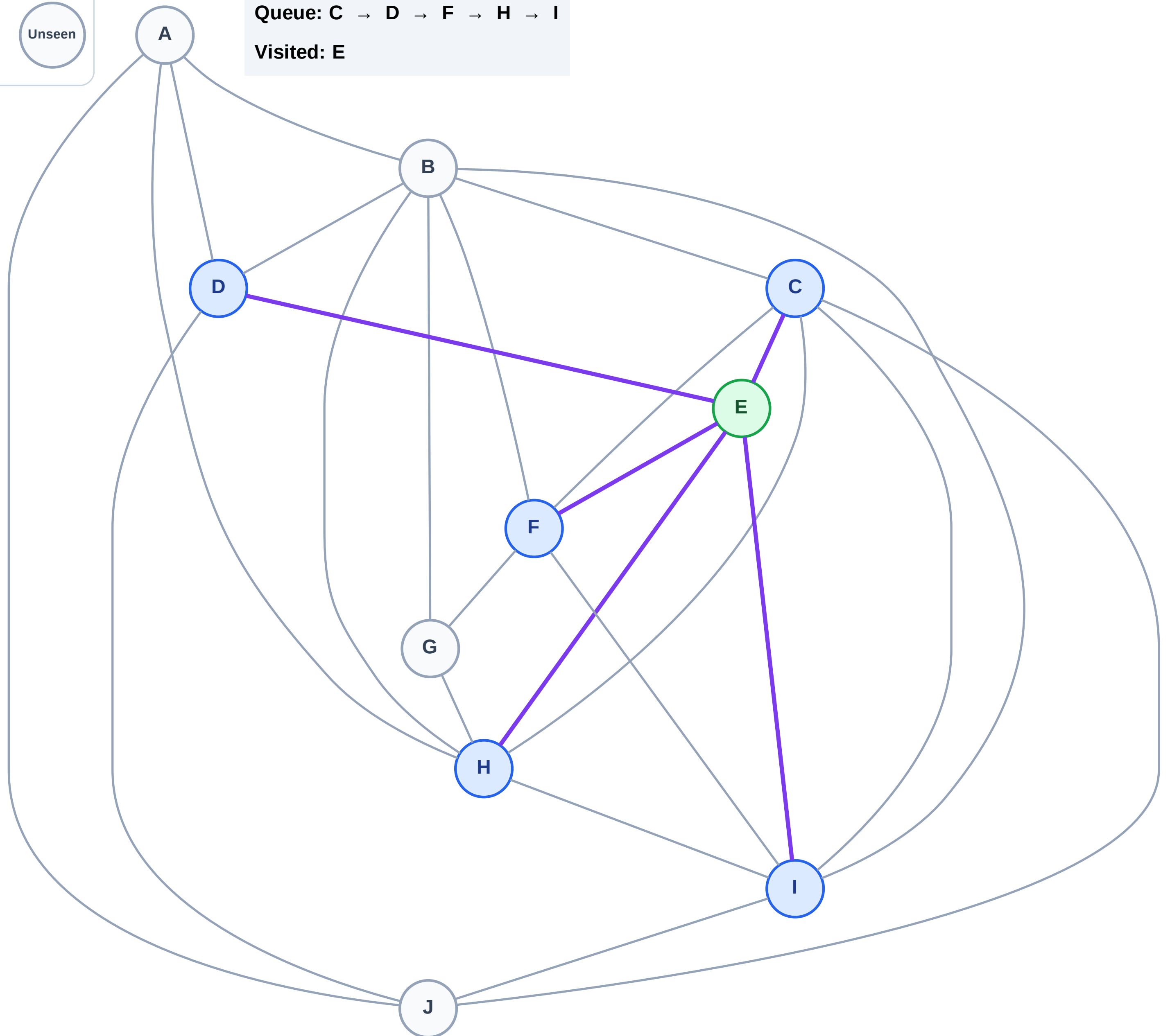


BFS Traversal

Step 3: Discover C, D, F, H, I from E



Queue: C → D → F → H → I
Visited: E



BFS Traversal

Step 4: Process node C

Legend

Complete

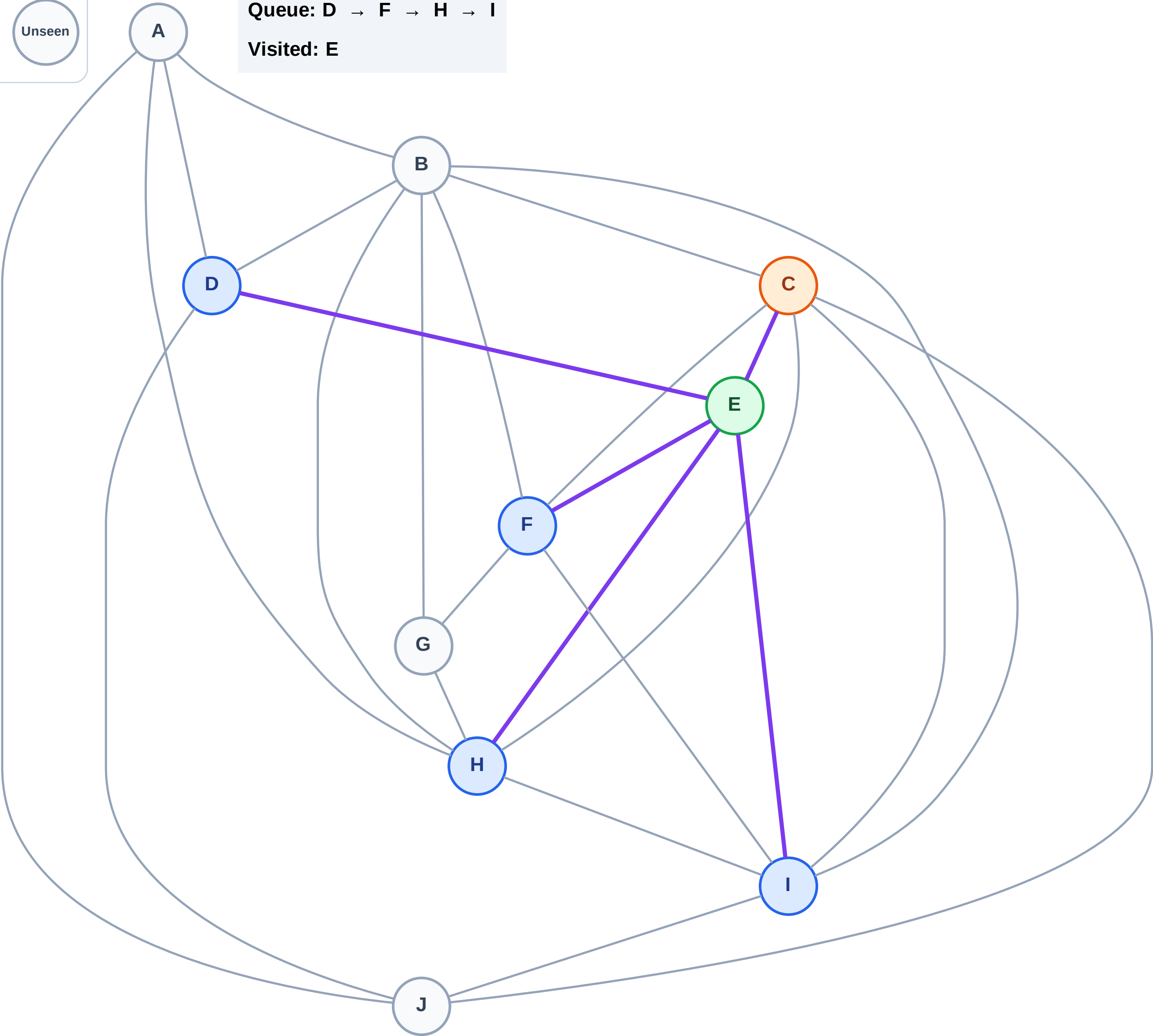
Processing

Discovered

Unseen

Queue: D → F → H → I

Visited: E



BFS Traversal

Step 5: Discover B, J from C

Legend

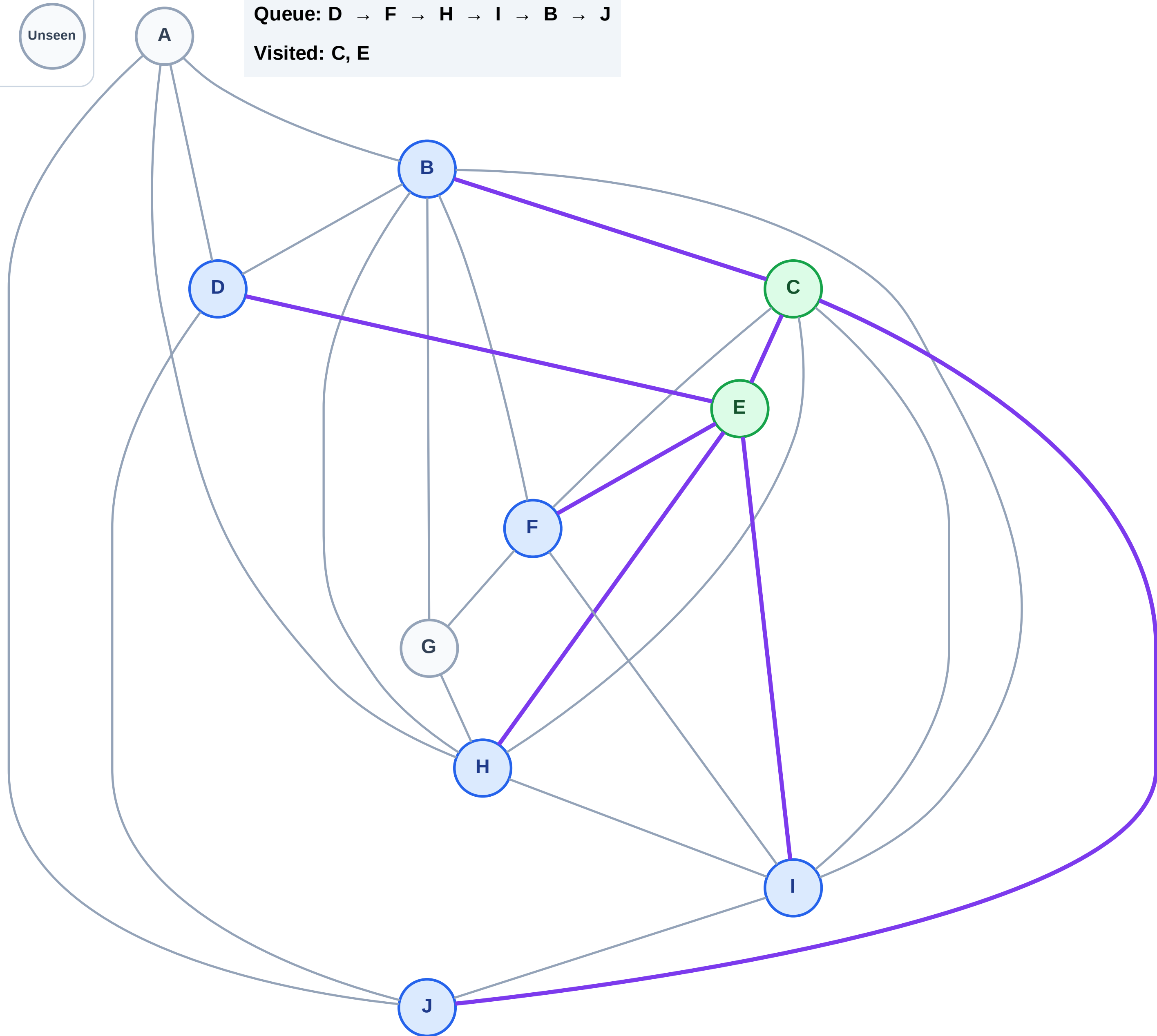
Complete

Processing

Discovered

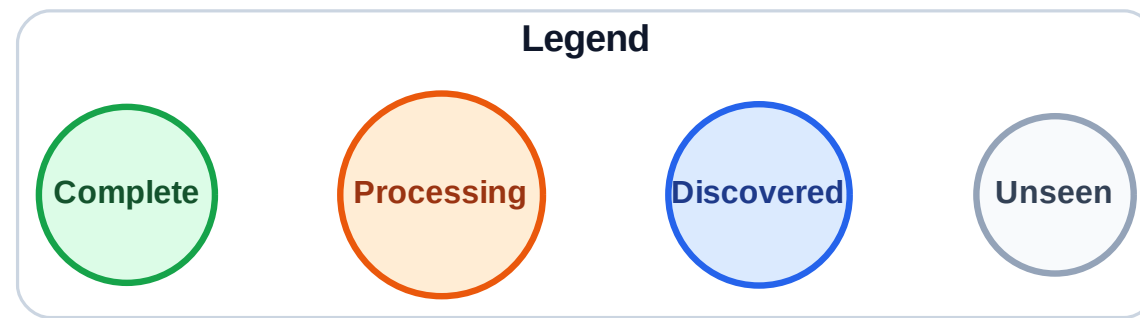
Unseen

Queue: D → F → H → I → B → J
Visited: C, E

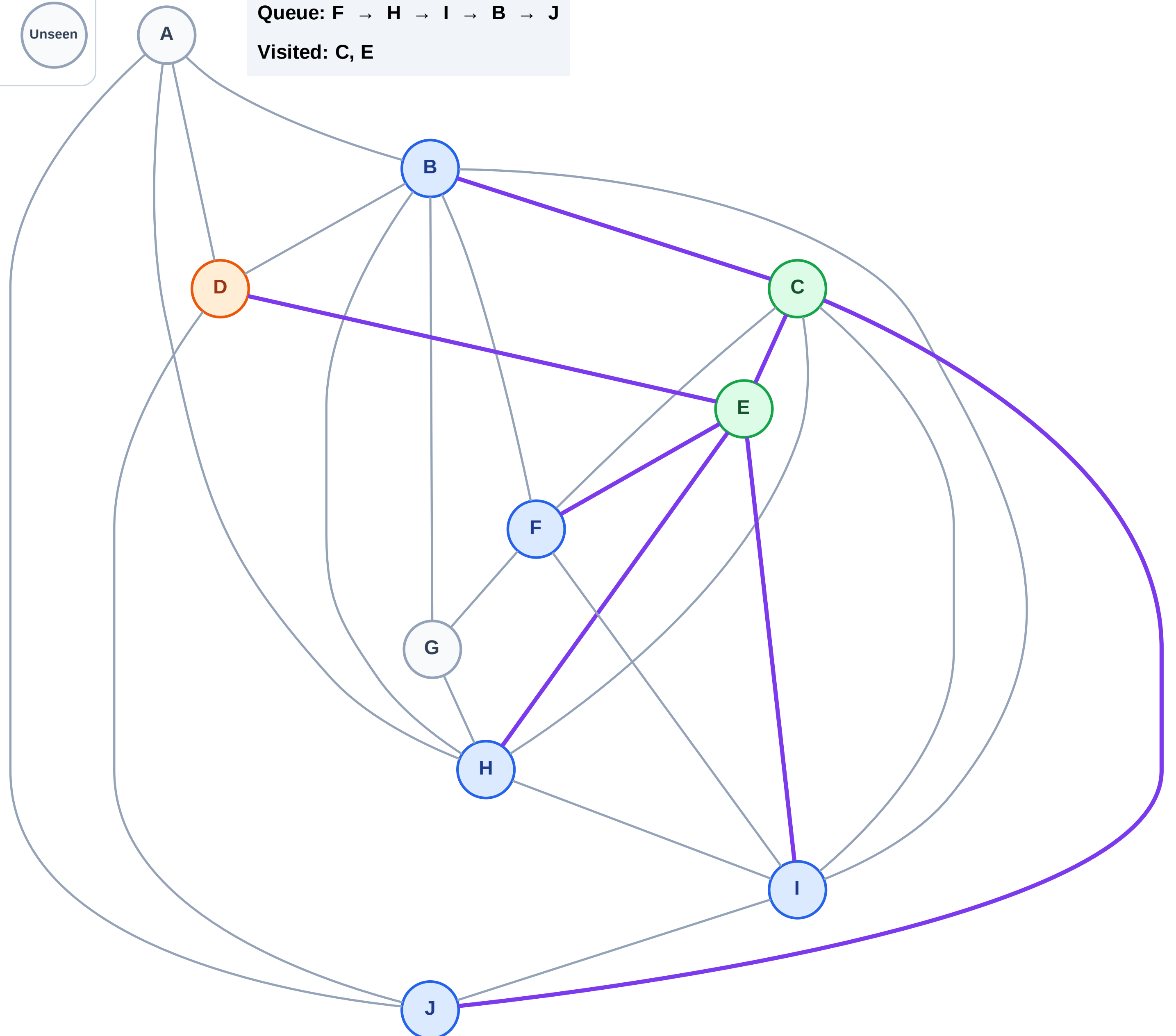


BFS Traversal

Step 6: Process node D

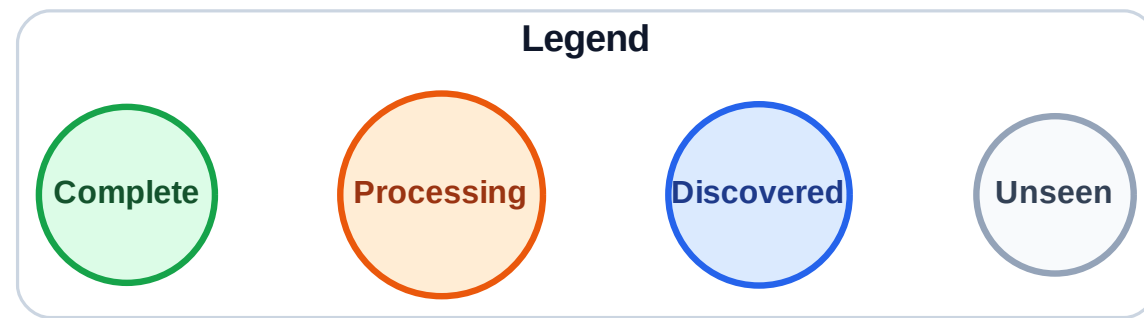


Queue: F → H → I → B → J
Visited: C, E

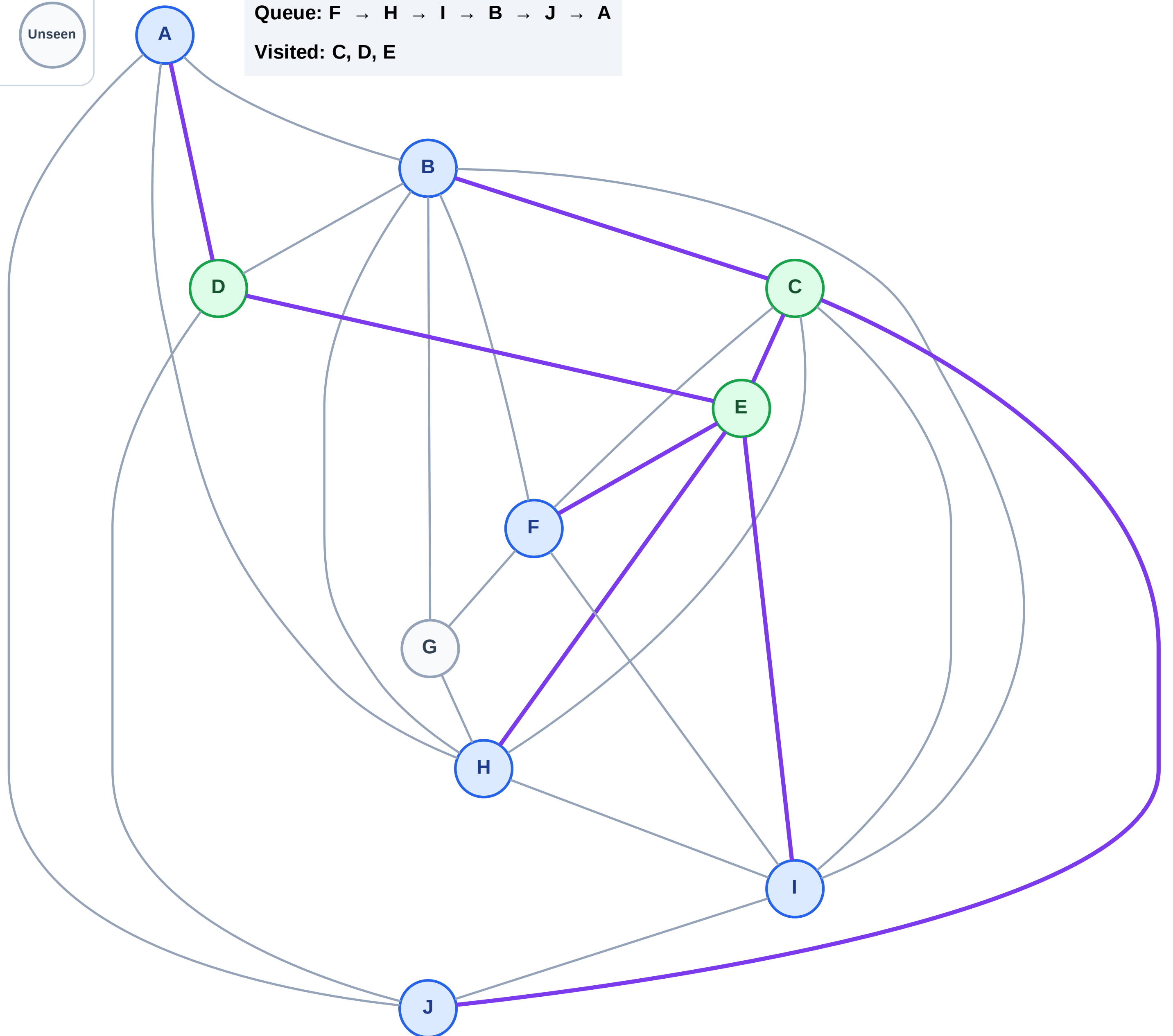


BFS Traversal

Step 7: Discover A from D



Queue: F → H → I → B → J → A
Visited: C, D, E



BFS Traversal

Step 8: Process node F

Legend

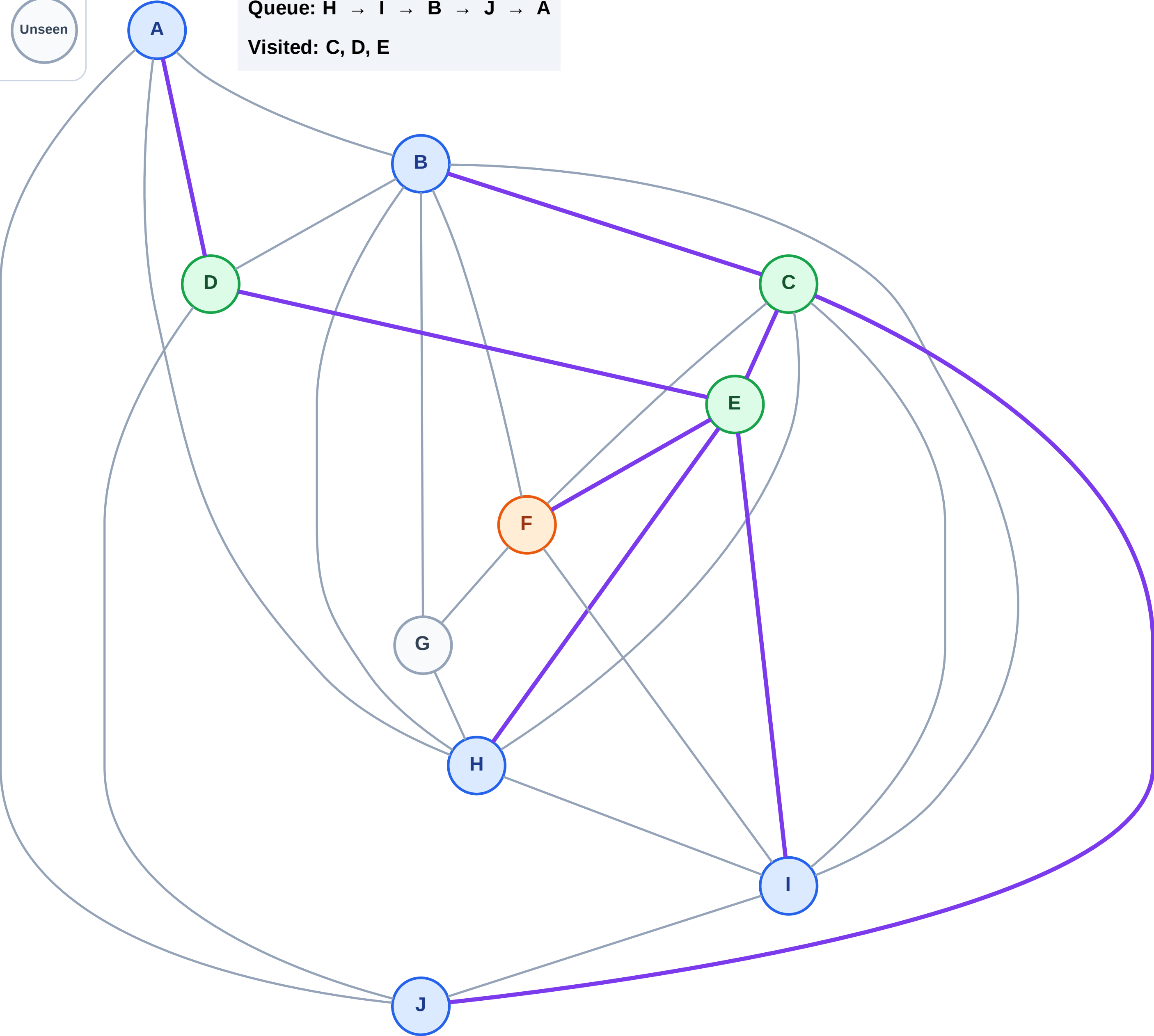
Complete

Processing

Discovered

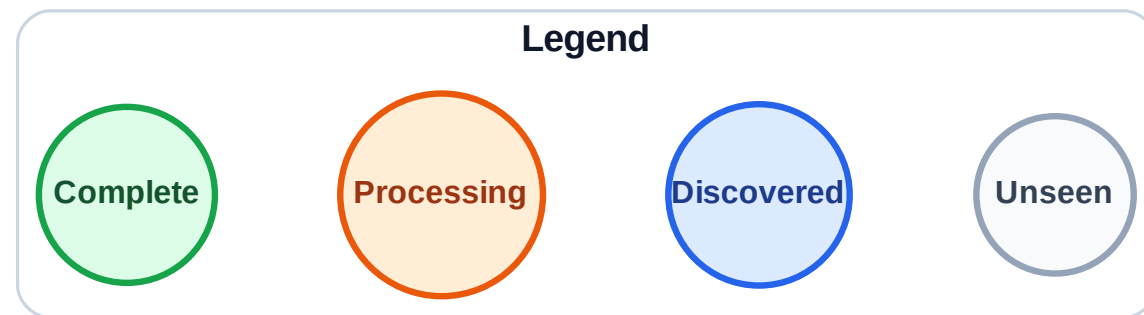
Unseen

Queue: H → I → B → J → A
Visited: C, D, E

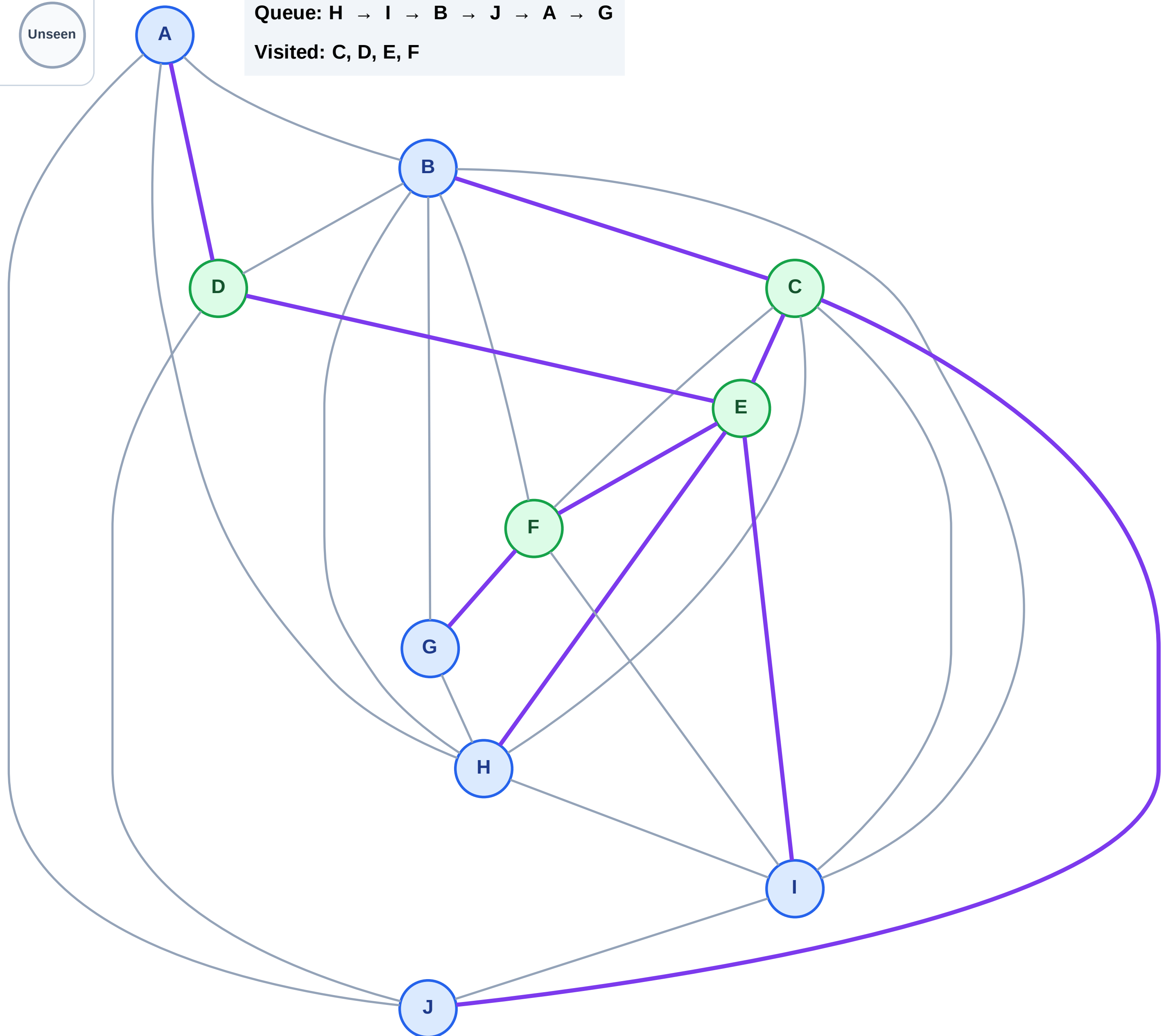


BFS Traversal

Step 9: Discover G from F

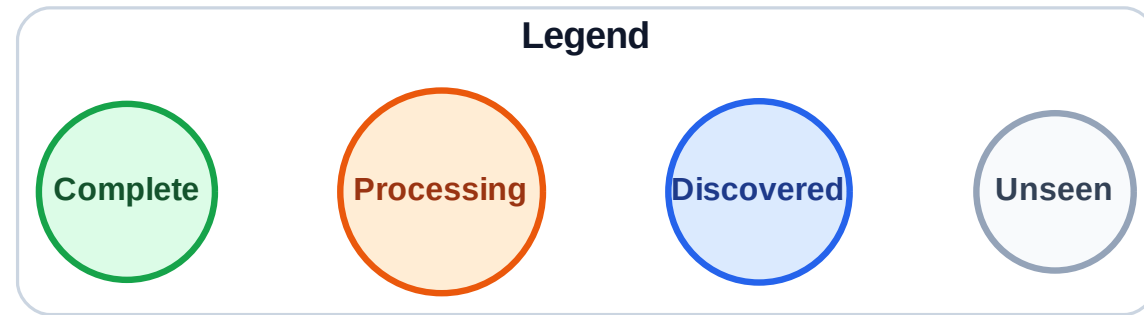


Queue: H → I → B → J → A → G
Visited: C, D, E, F

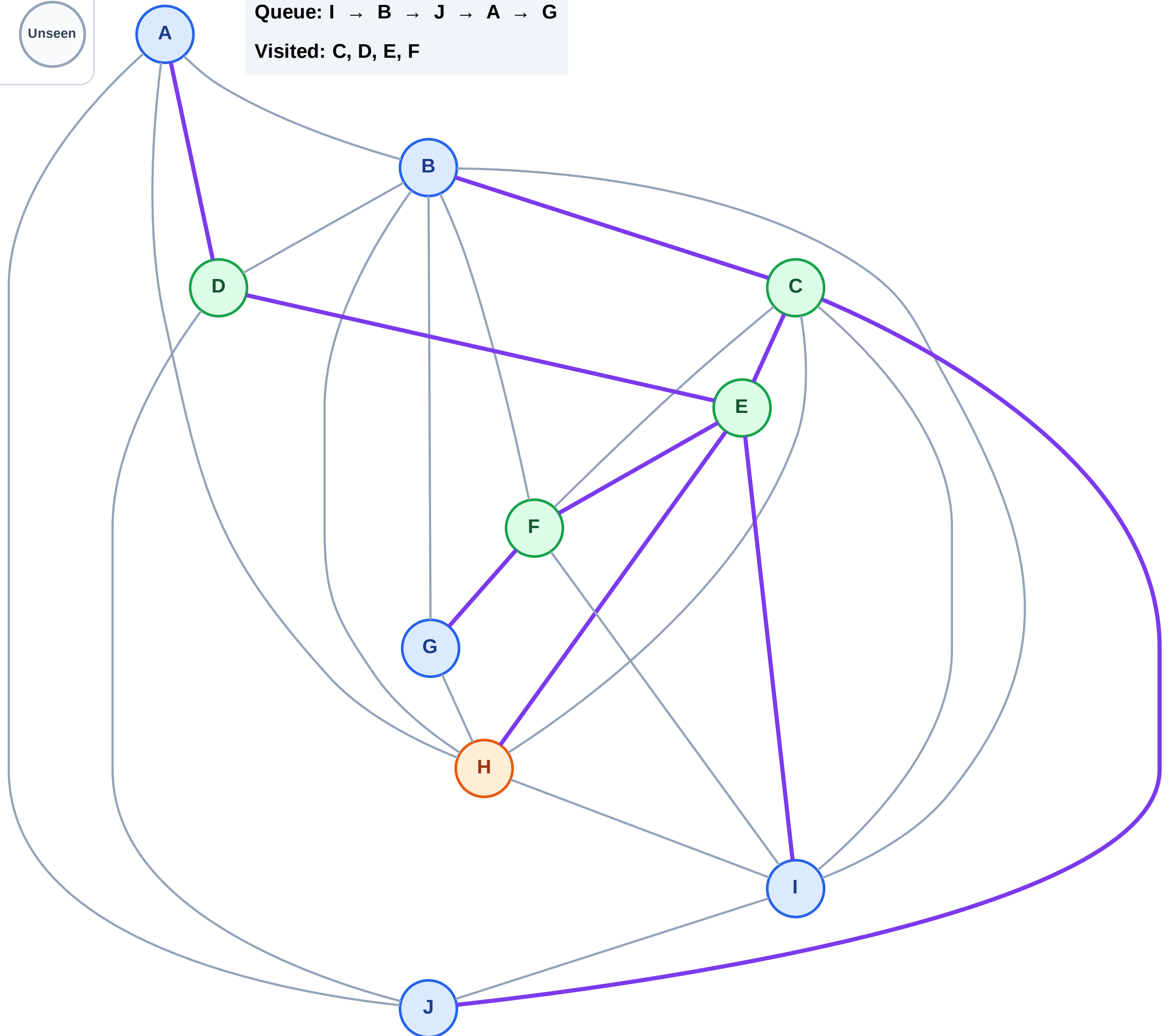


BFS Traversal

Step 10: Process node H

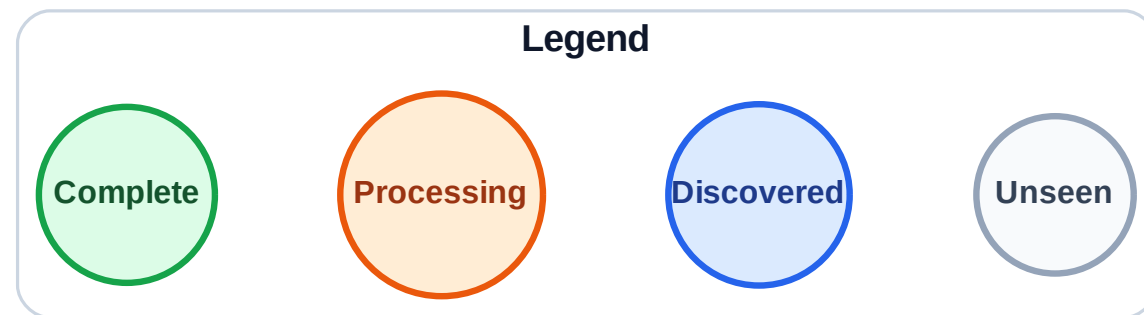


Queue: I → B → J → A → G
Visited: C, D, E, F



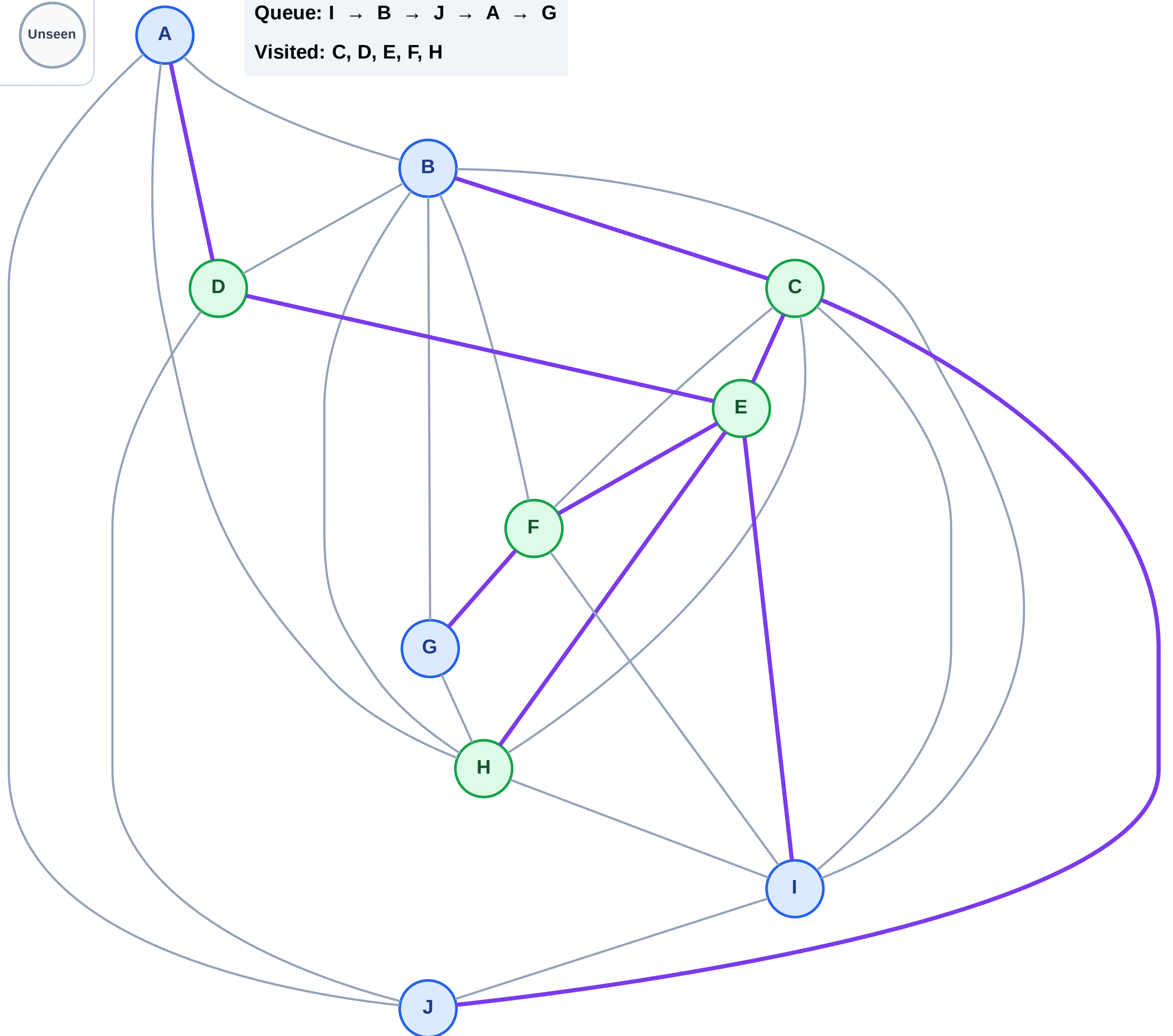
BFS Traversal

Step 11: No new neighbors from H



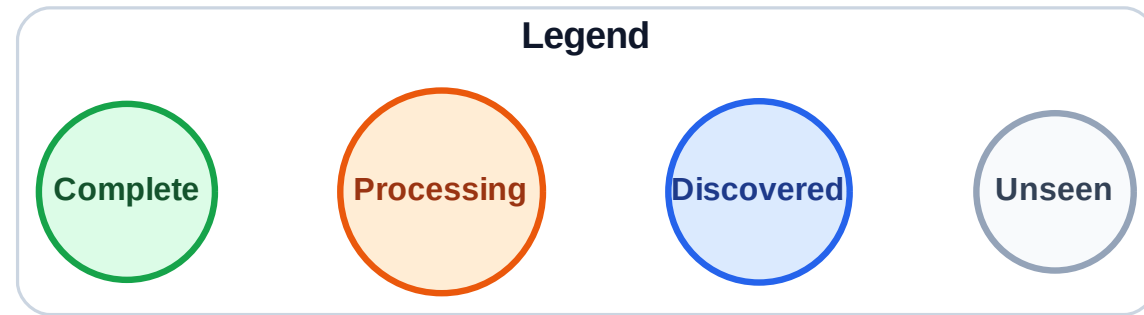
Queue: I → B → J → A → G

Visited: C, D, E, F, H



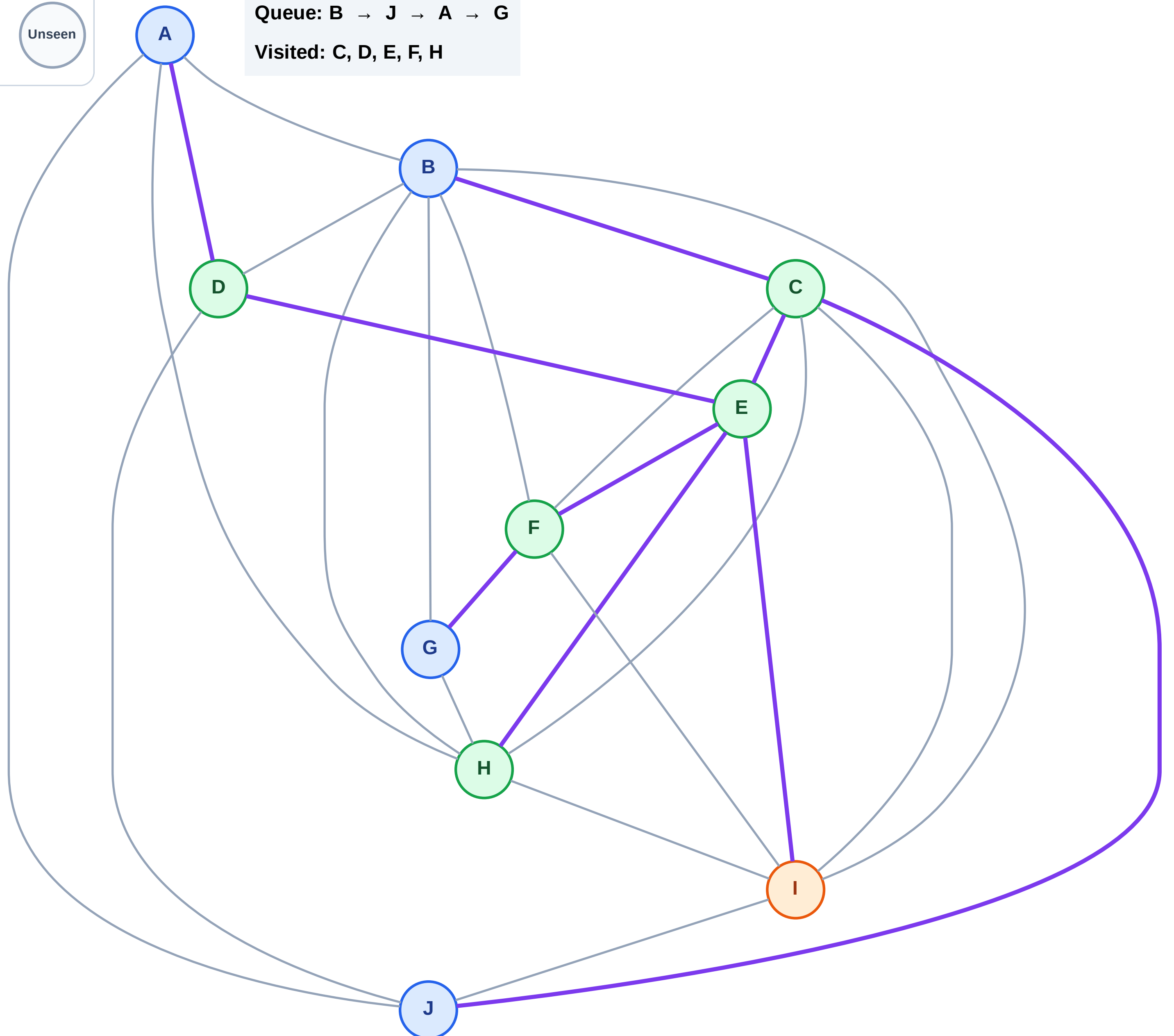
BFS Traversal

Step 12: Process node I



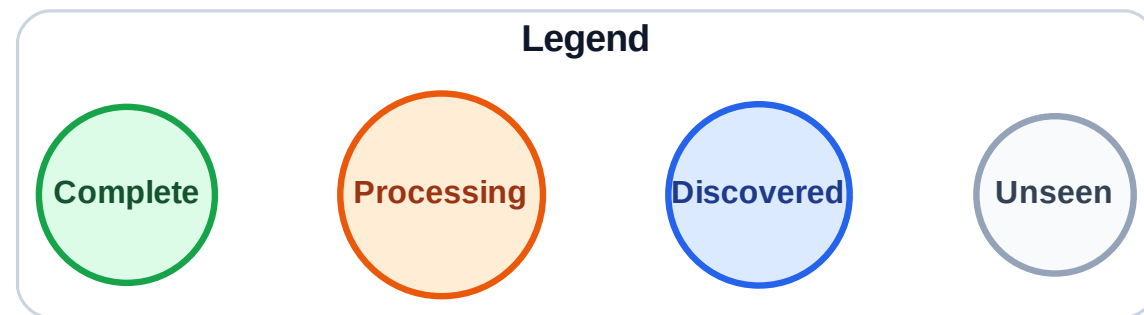
Queue: B → J → A → G

Visited: C, D, E, F, H

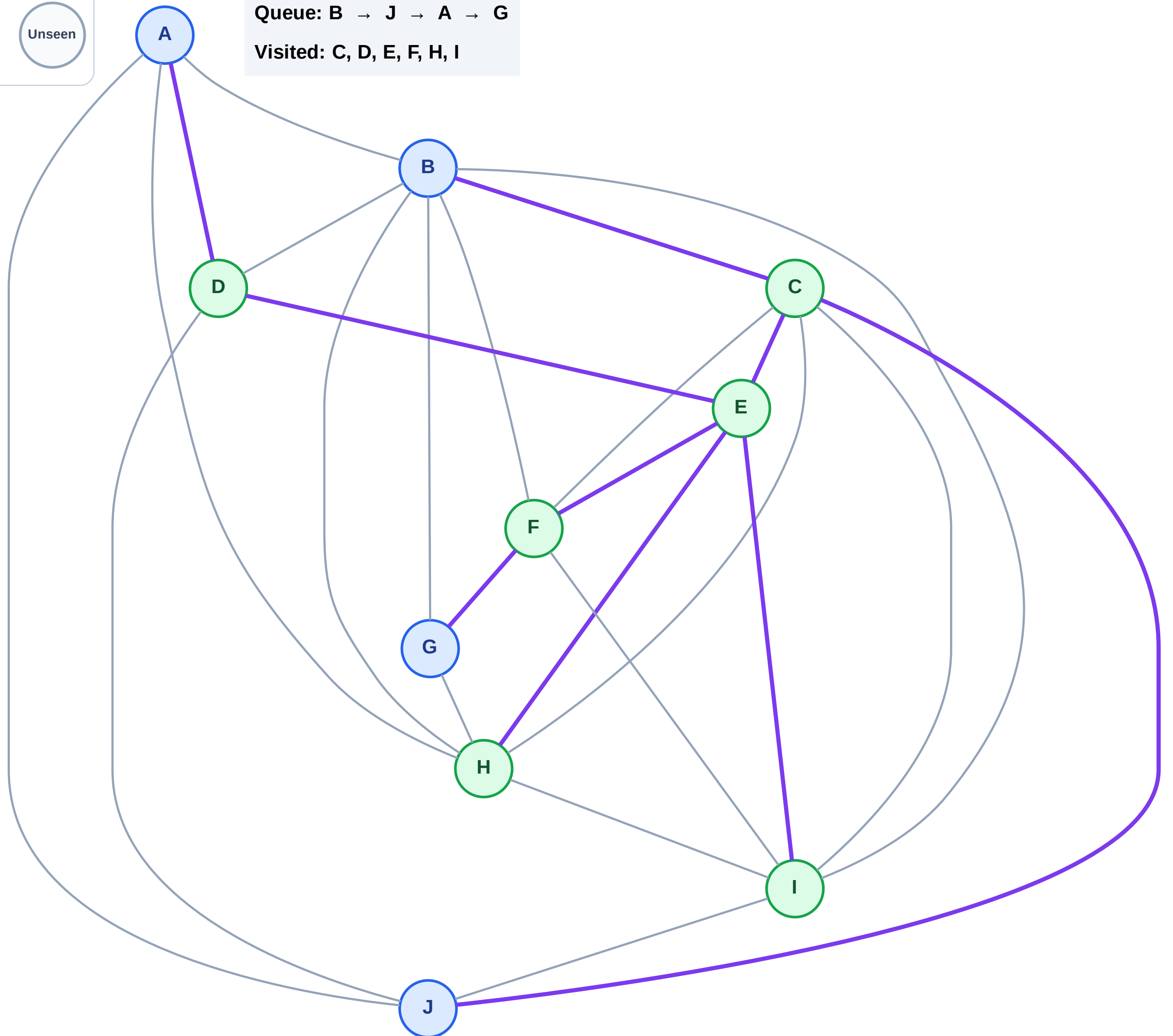


BFS Traversal

Step 13: No new neighbors from I

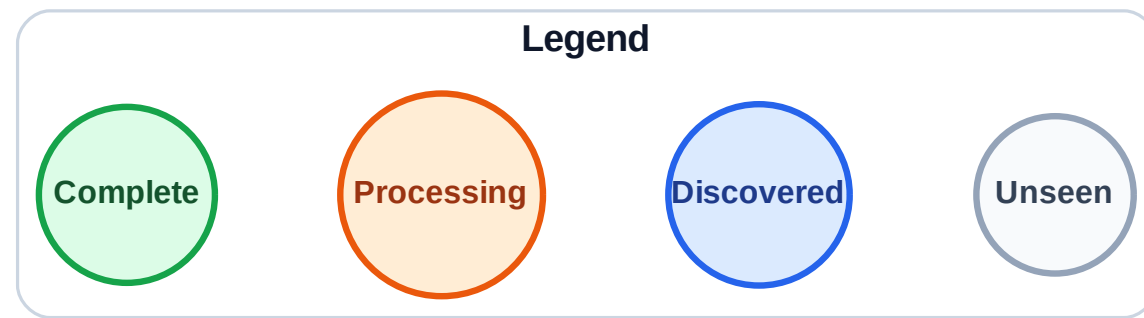


Queue: B → J → A → G
Visited: C, D, E, F, H, I



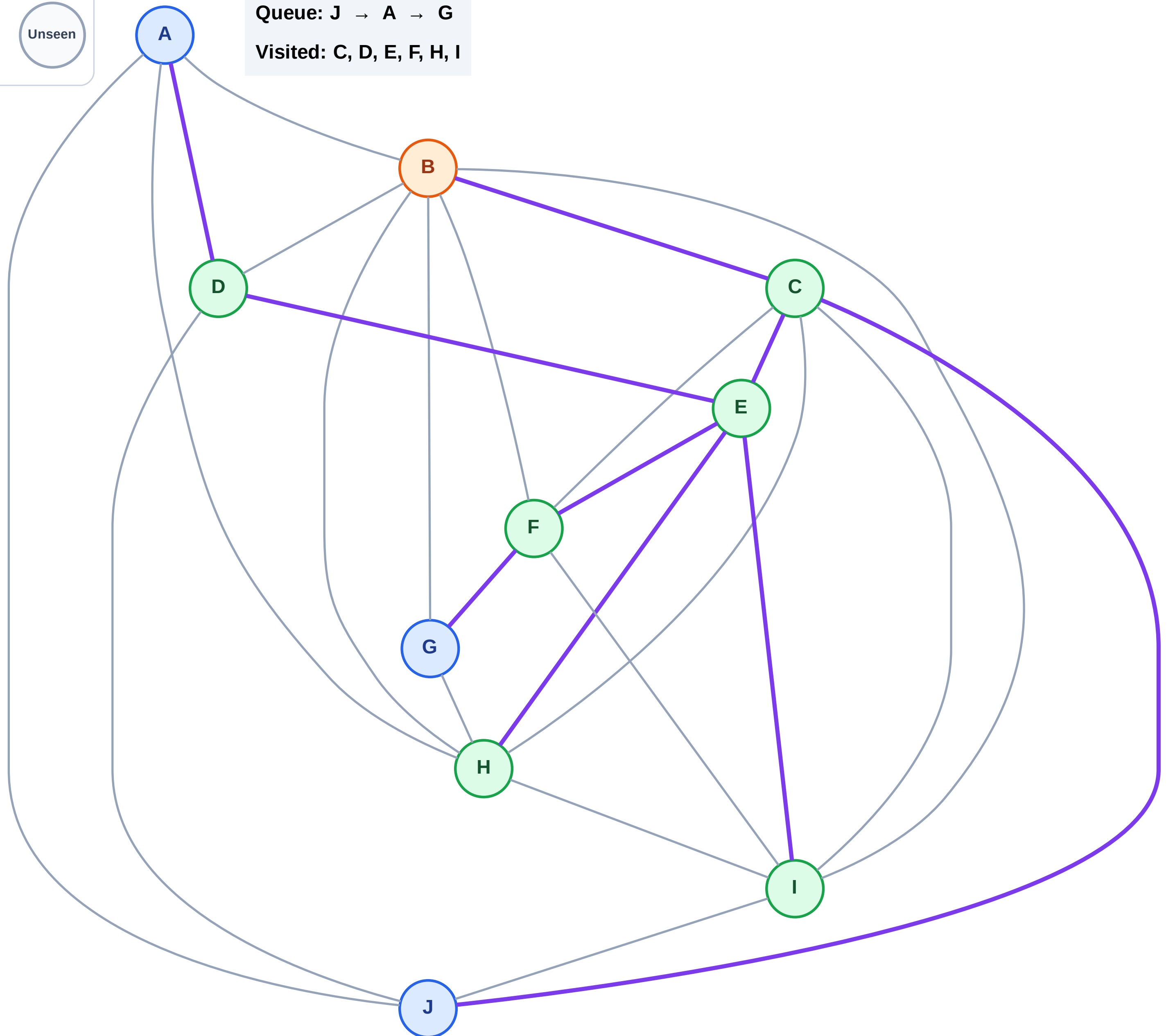
BFS Traversal

Step 14: Process node B



Queue: J → A → G

Visited: C, D, E, F, H, I

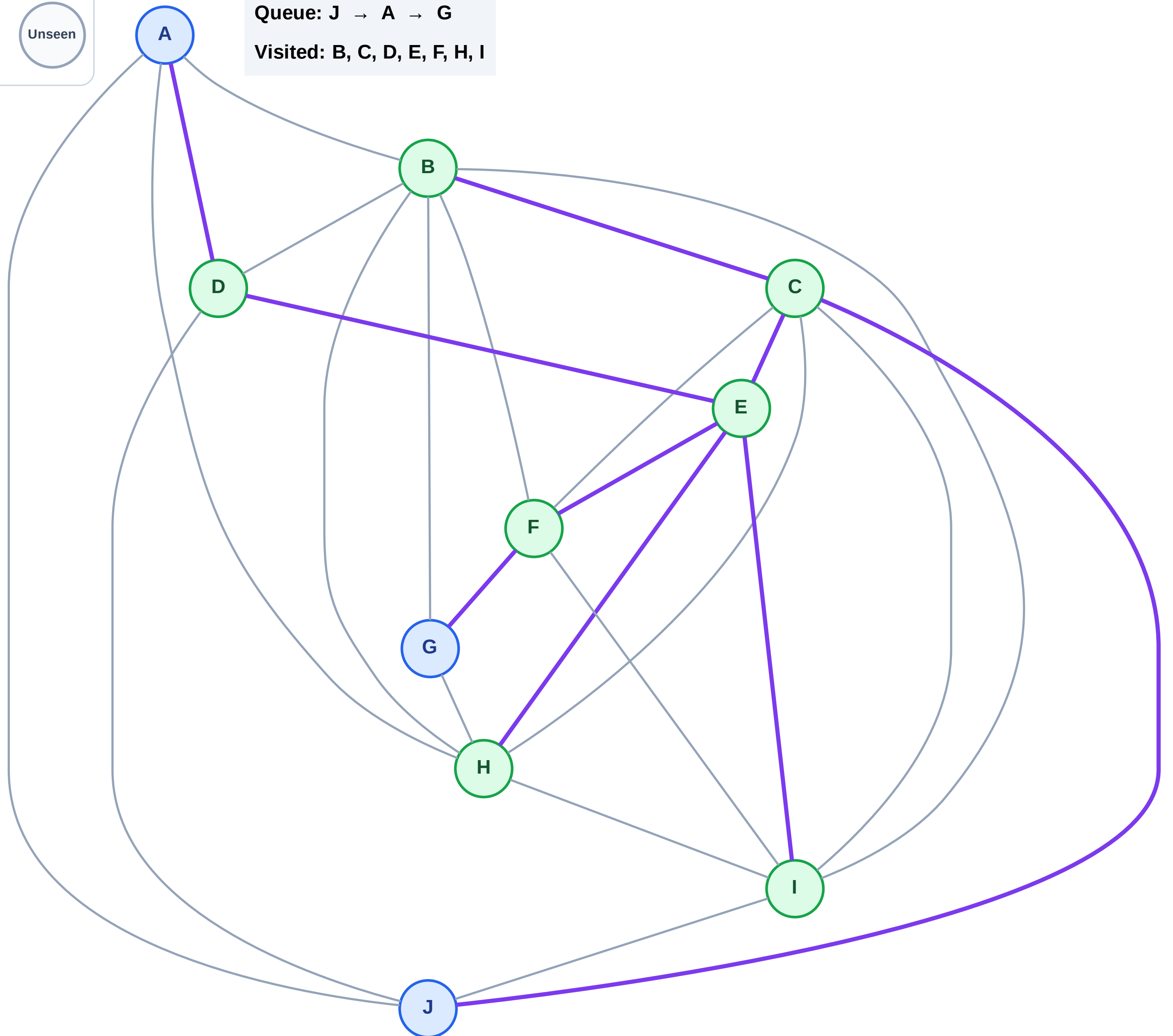
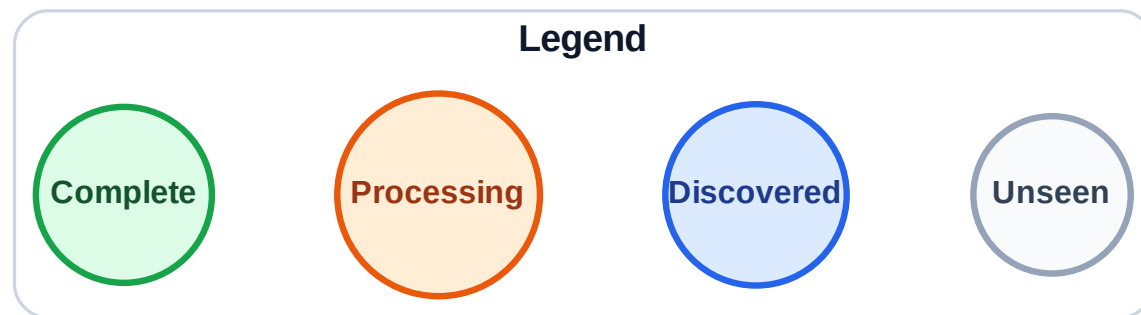


BFS Traversal

Step 15: No new neighbors from B

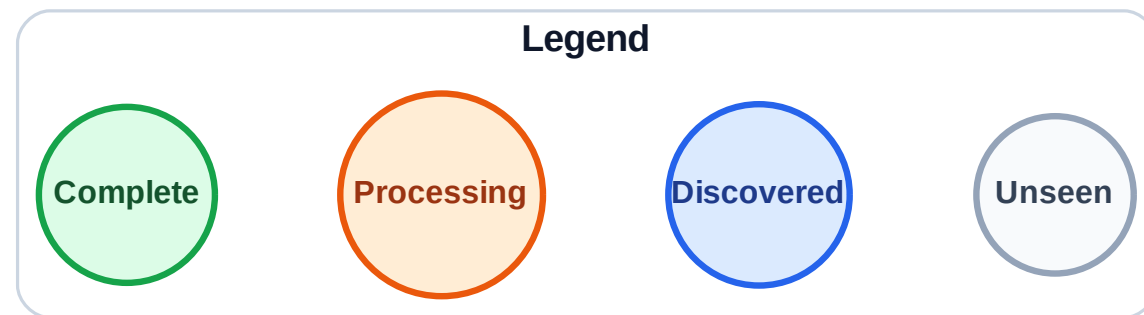
Queue: J → A → G

Visited: B, C, D, E, F, H, I

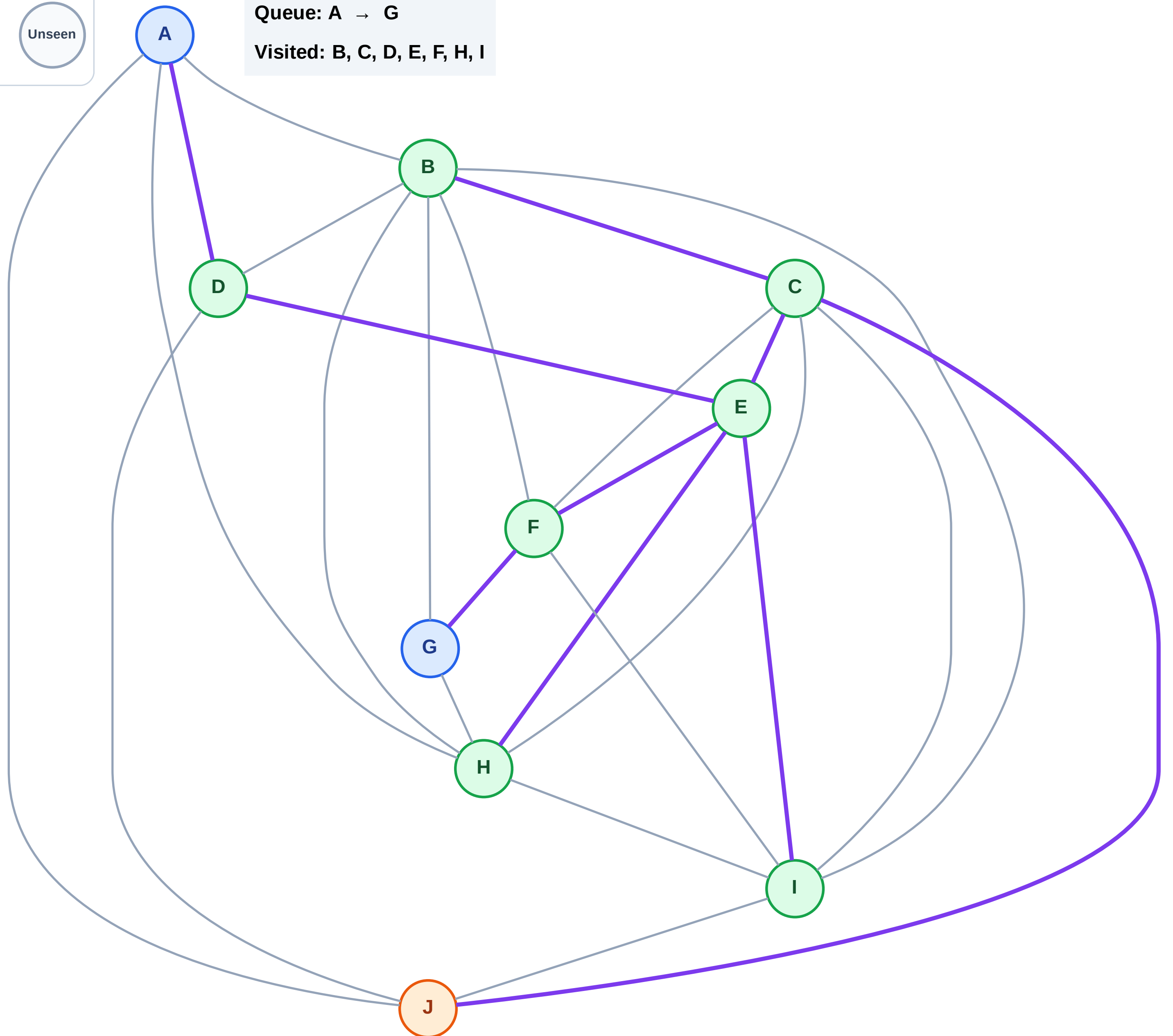


BFS Traversal

Step 16: Process node J

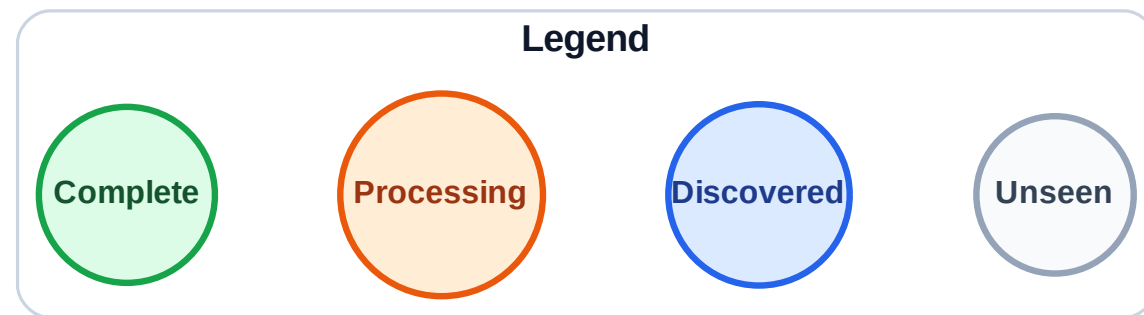


Queue: A → G
Visited: B, C, D, E, F, H, I



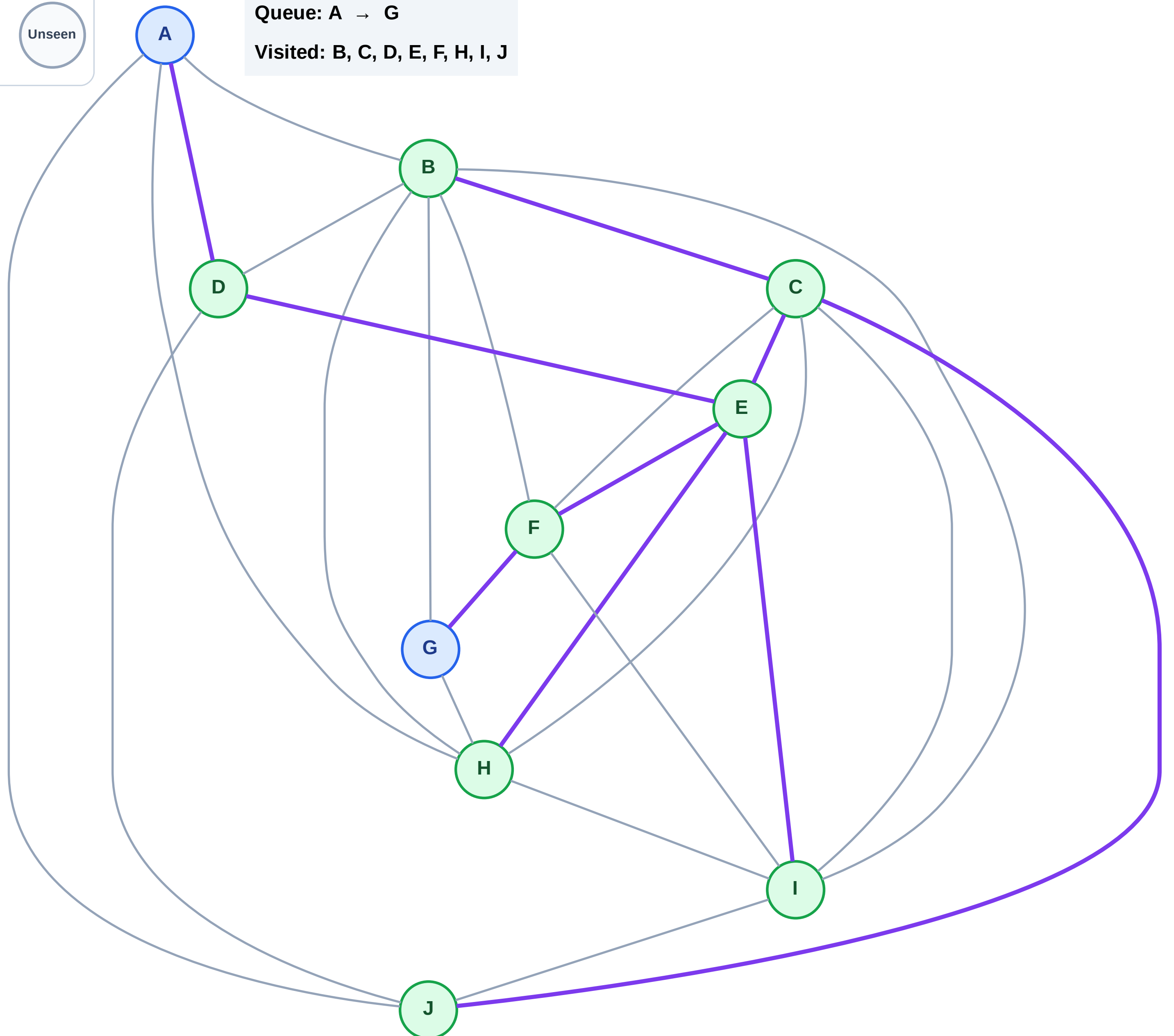
BFS Traversal

Step 17: No new neighbors from J



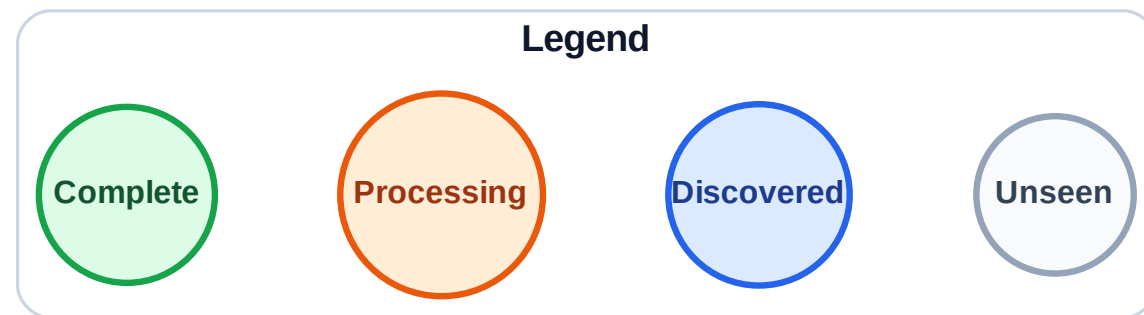
Queue: A → G

Visited: B, C, D, E, F, H, I, J



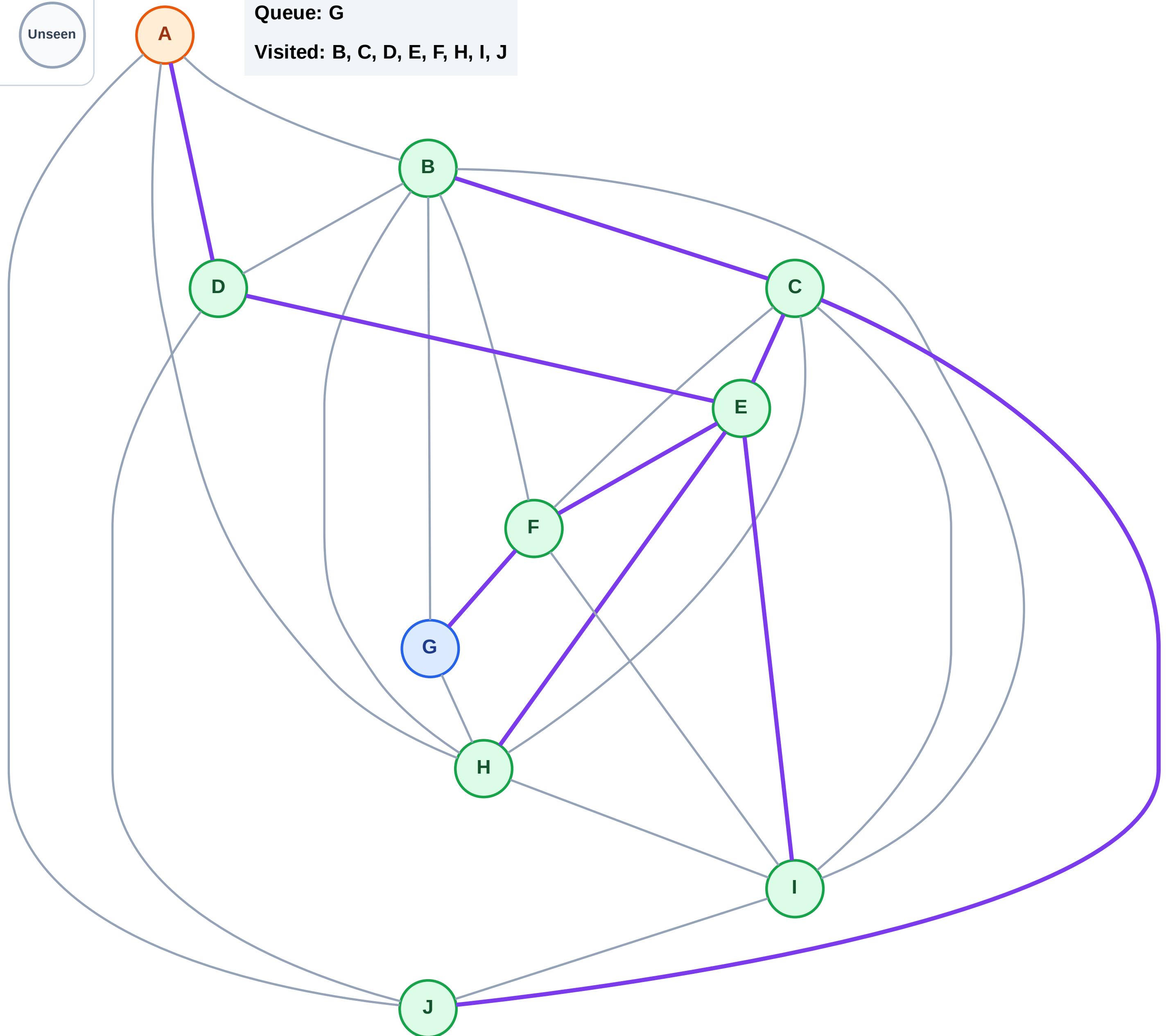
BFS Traversal

Step 18: Process node A



Queue: G

Visited: B, C, D, E, F, H, I, J



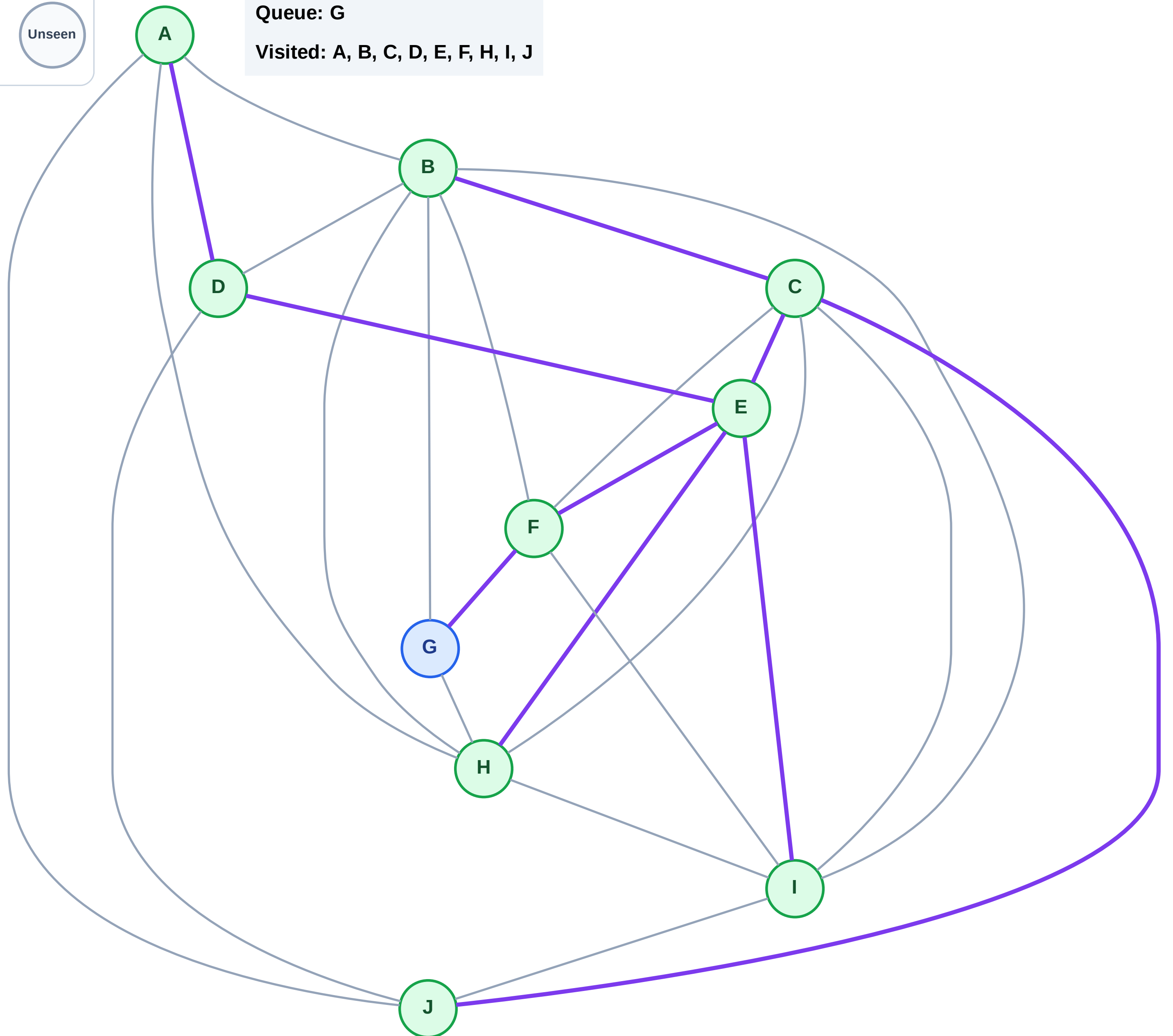
BFS Traversal

Step 19: No new neighbors from A

Queue: G

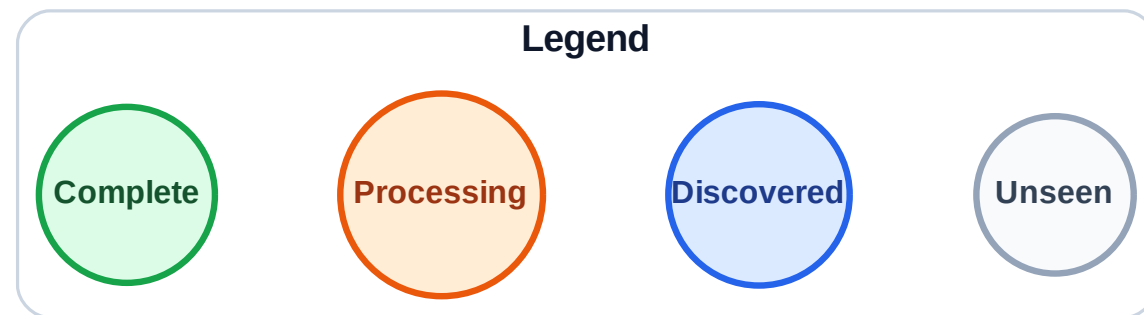
Visited: A, B, C, D, E, F, H, I, J

Legend



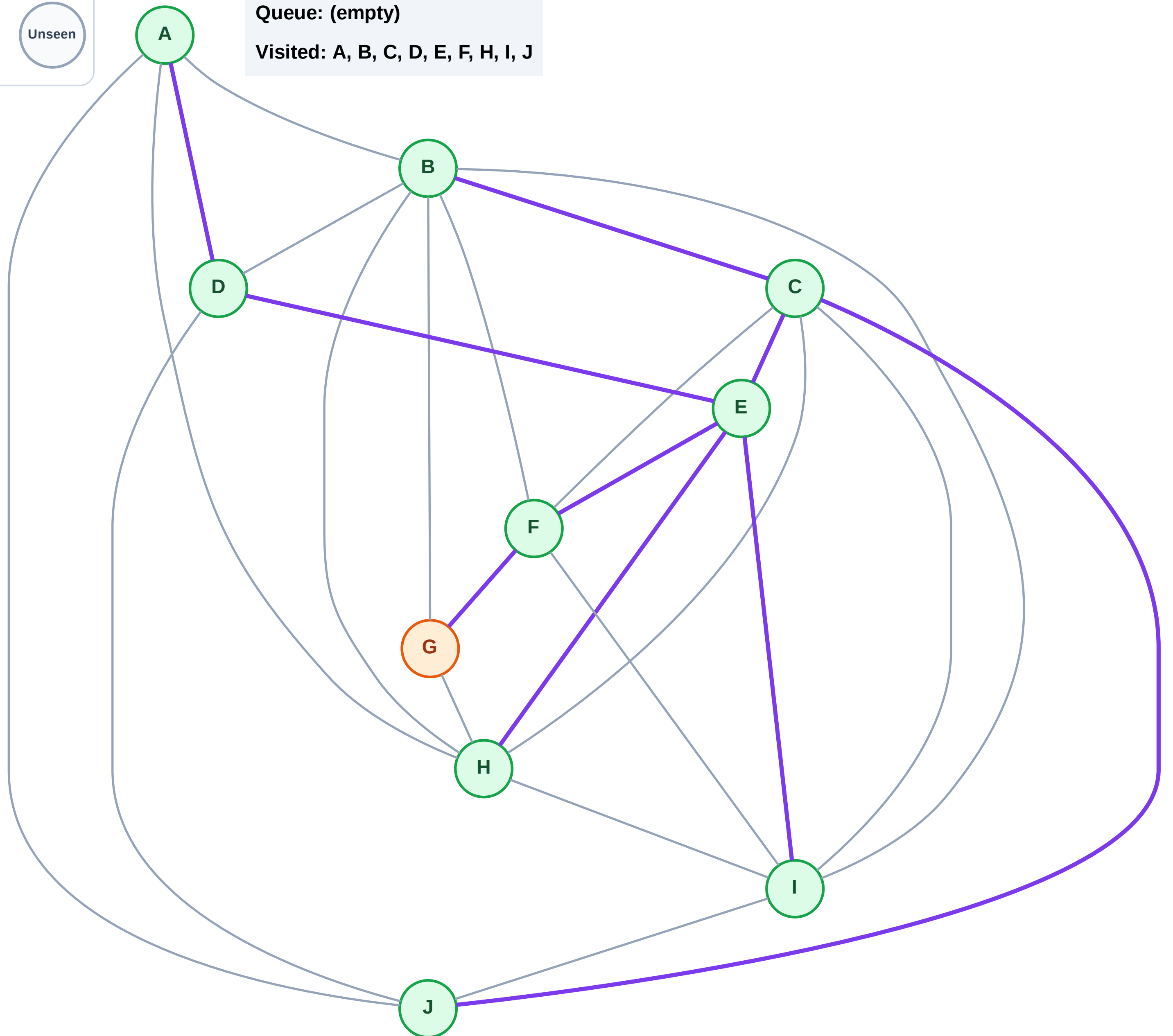
BFS Traversal

Step 20: Process node G



Queue: (empty)

Visited: A, B, C, D, E, F, H, I, J



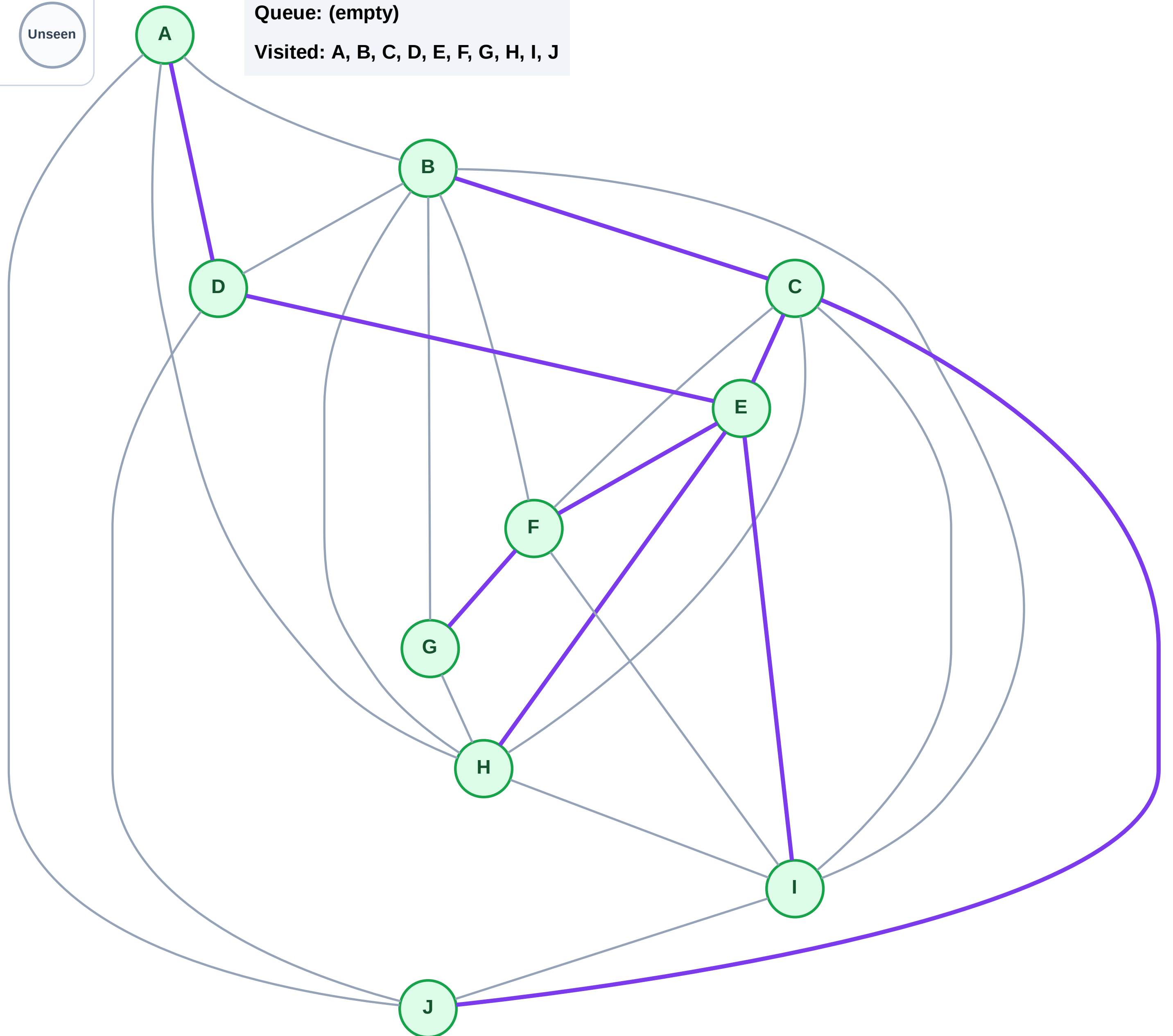
BFS Traversal

Step 21: No new neighbors from G

Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J

Legend



DFS Traversal

Step 1: Start at E

Legend

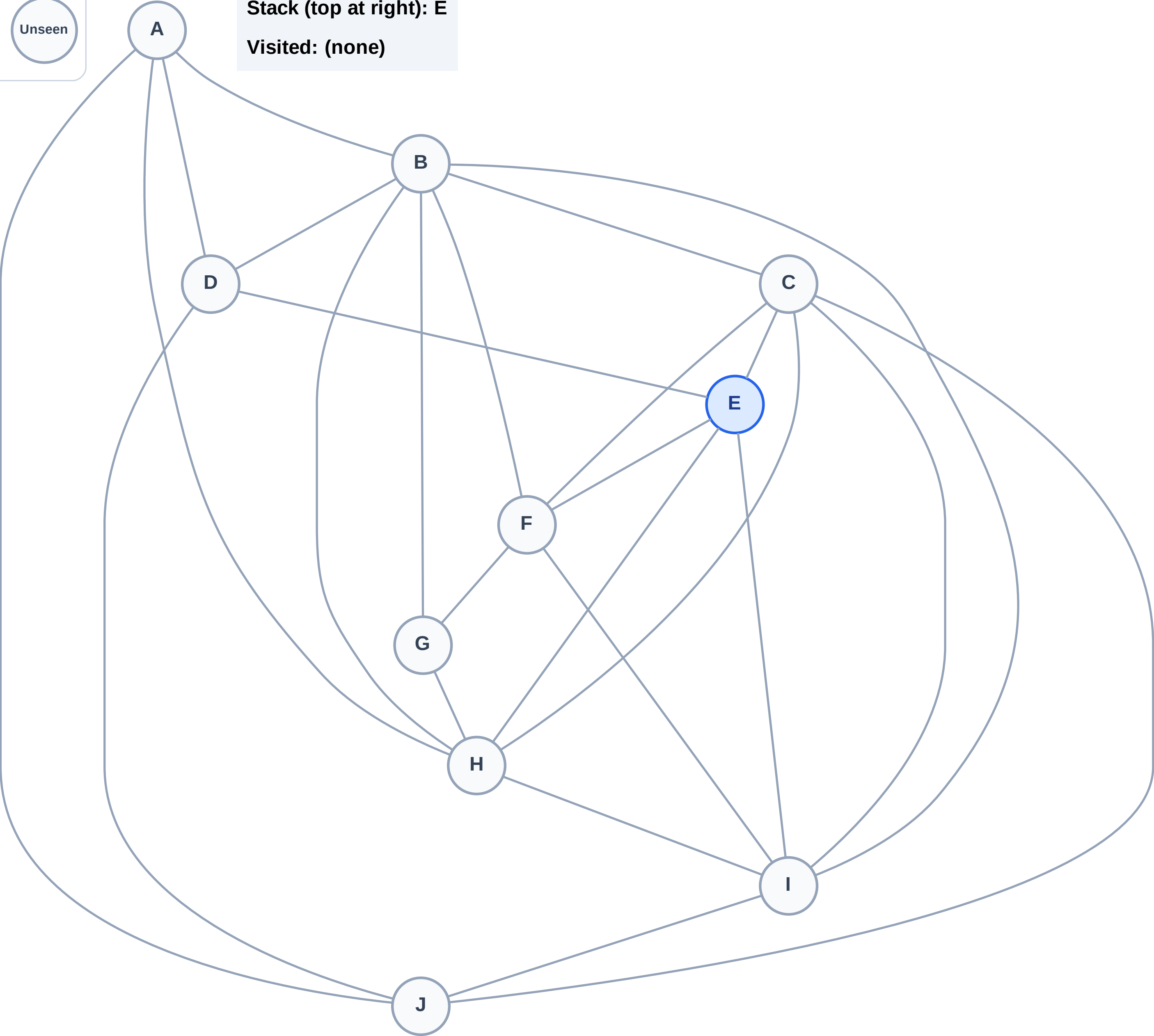
Complete

Processing

Discovered

Unseen

Stack (top at right): E
Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

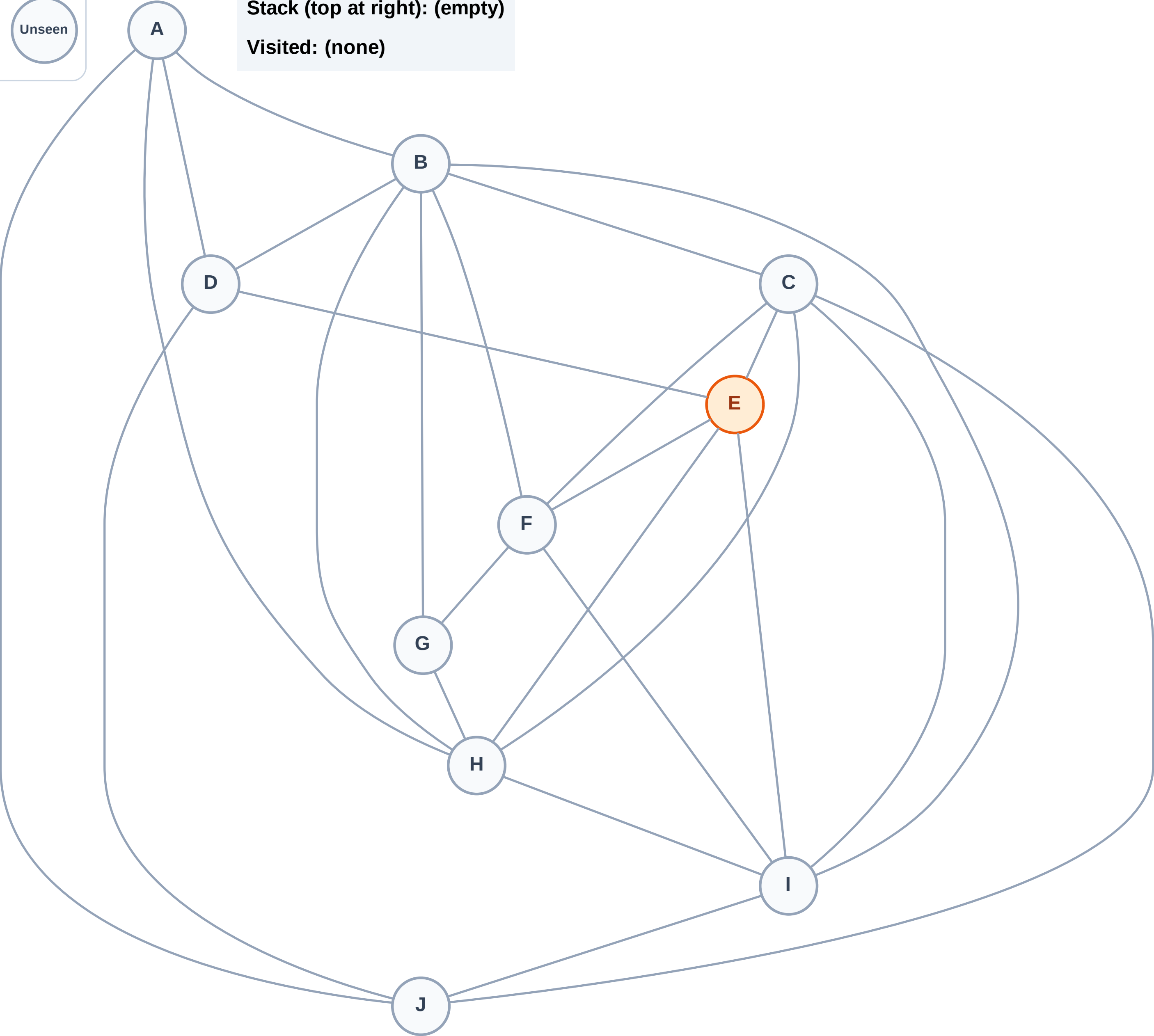
Complete

Processing

Discovered

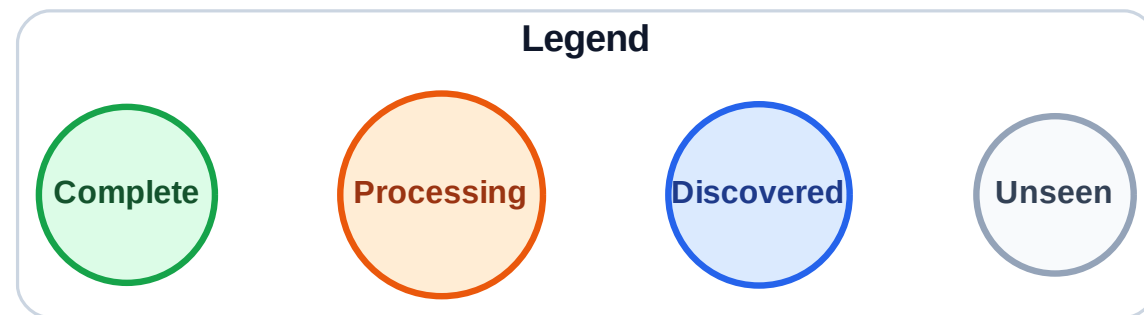
Unseen

Stack (top at right): (empty)
Visited: (none)

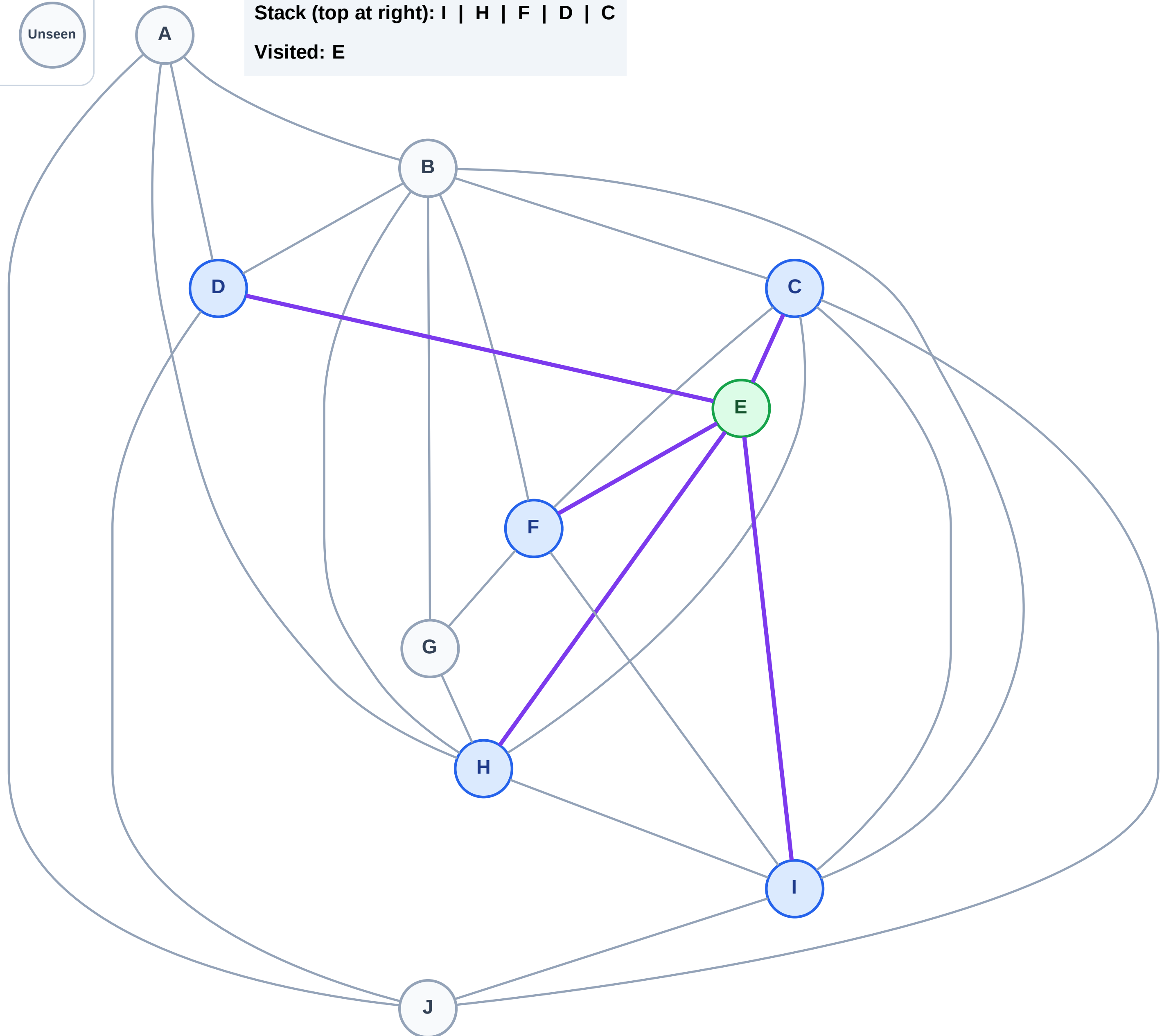


DFS Traversal

Step 3: Discover I, H, F, D, C from E



Stack (top at right): I | H | F | D | C
Visited: E



DFS Traversal

Step 4: Process node C

Legend

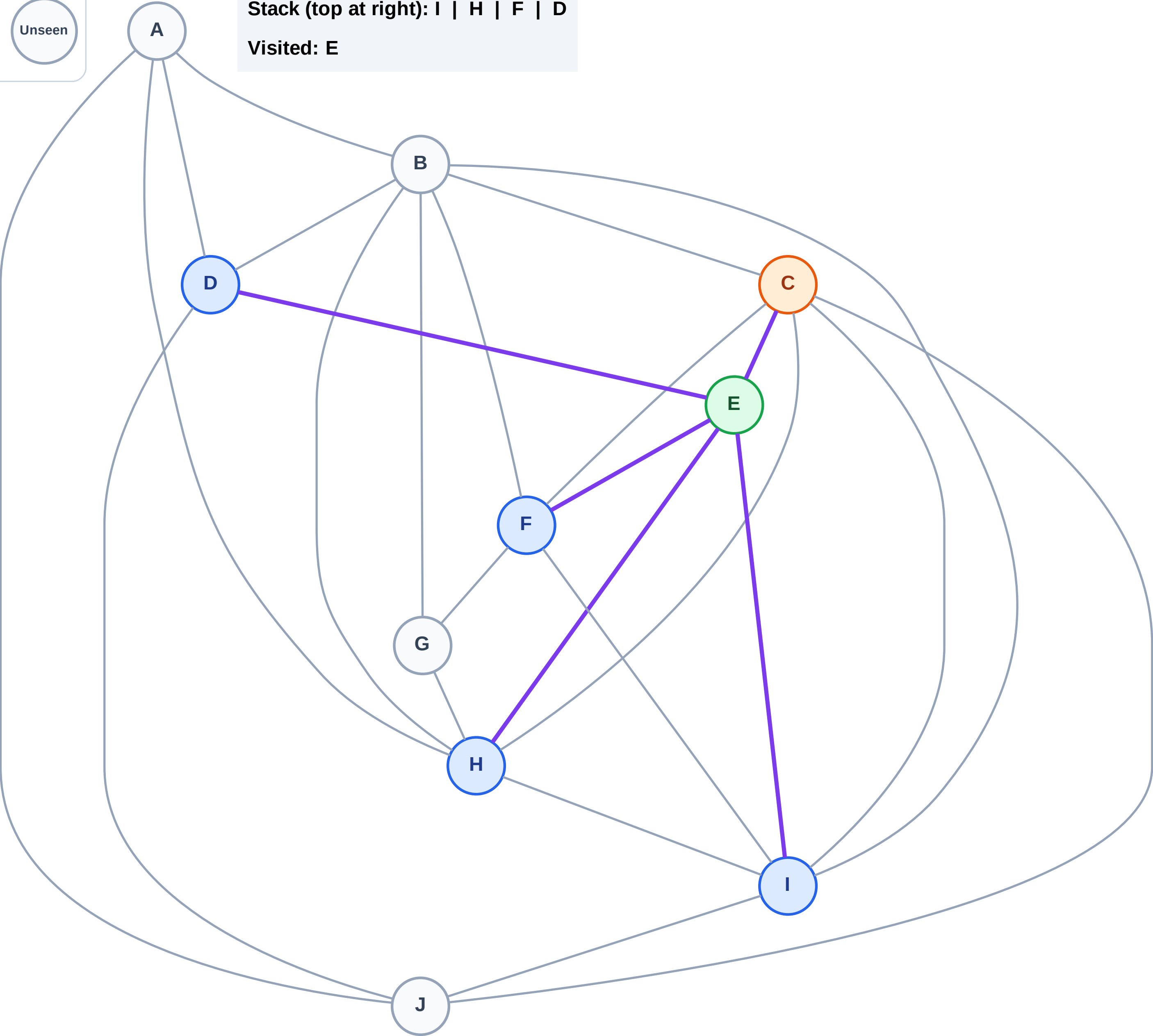
Complete

Processing

Discovered

Unseen

Stack (top at right): I | H | F | D
Visited: E



DFS Traversal

Step 5: Discover J, B from C

Legend

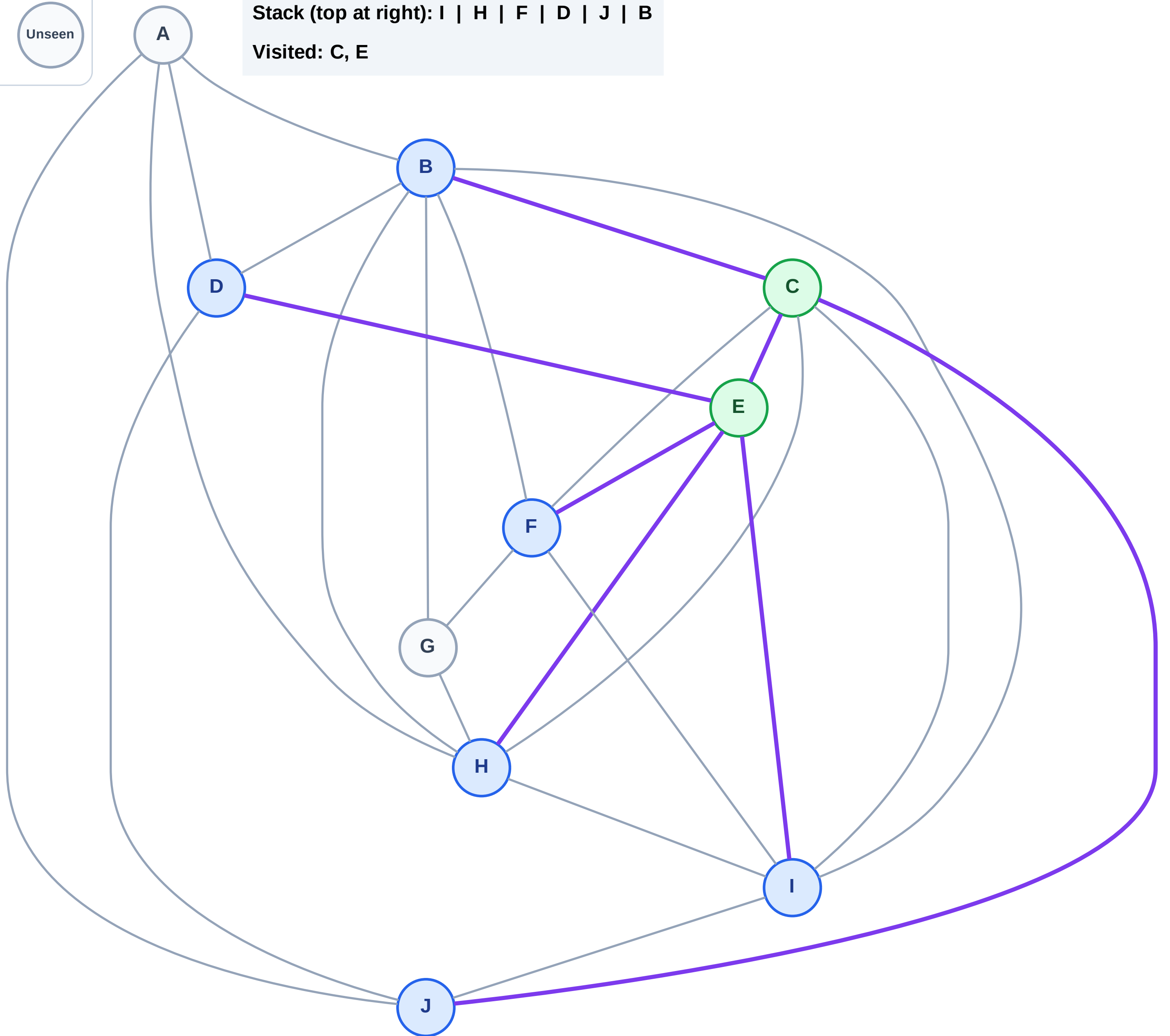
Complete

Processing

Discovered

Unseen

Stack (top at right): I | H | F | D | J | B
Visited: C, E



DFS Traversal

Step 6: Process node B

Legend

Complete

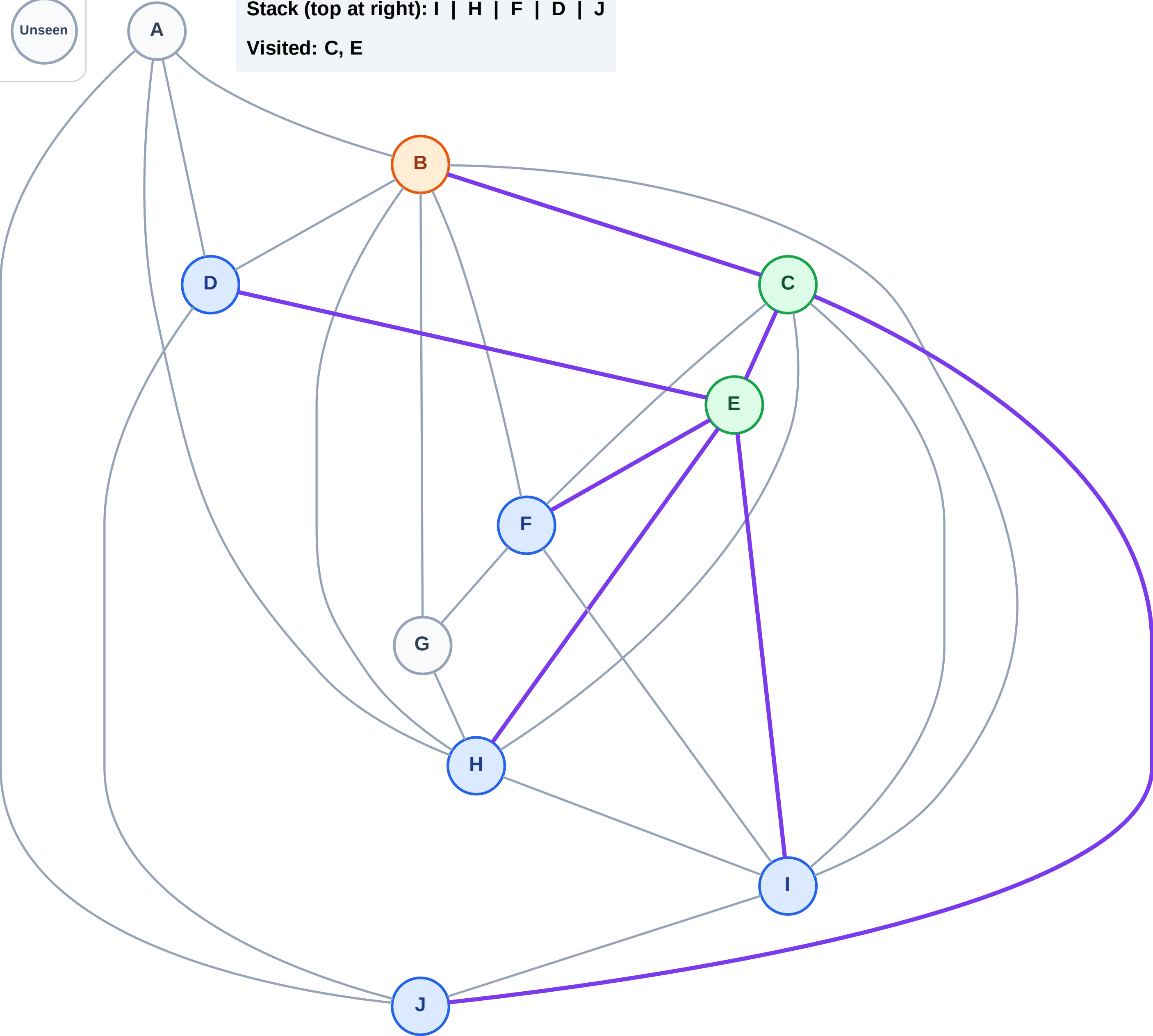
Processing

Discovered

Unseen

Stack (top at right): I | H | F | D | J

Visited: C, E



DFS Traversal

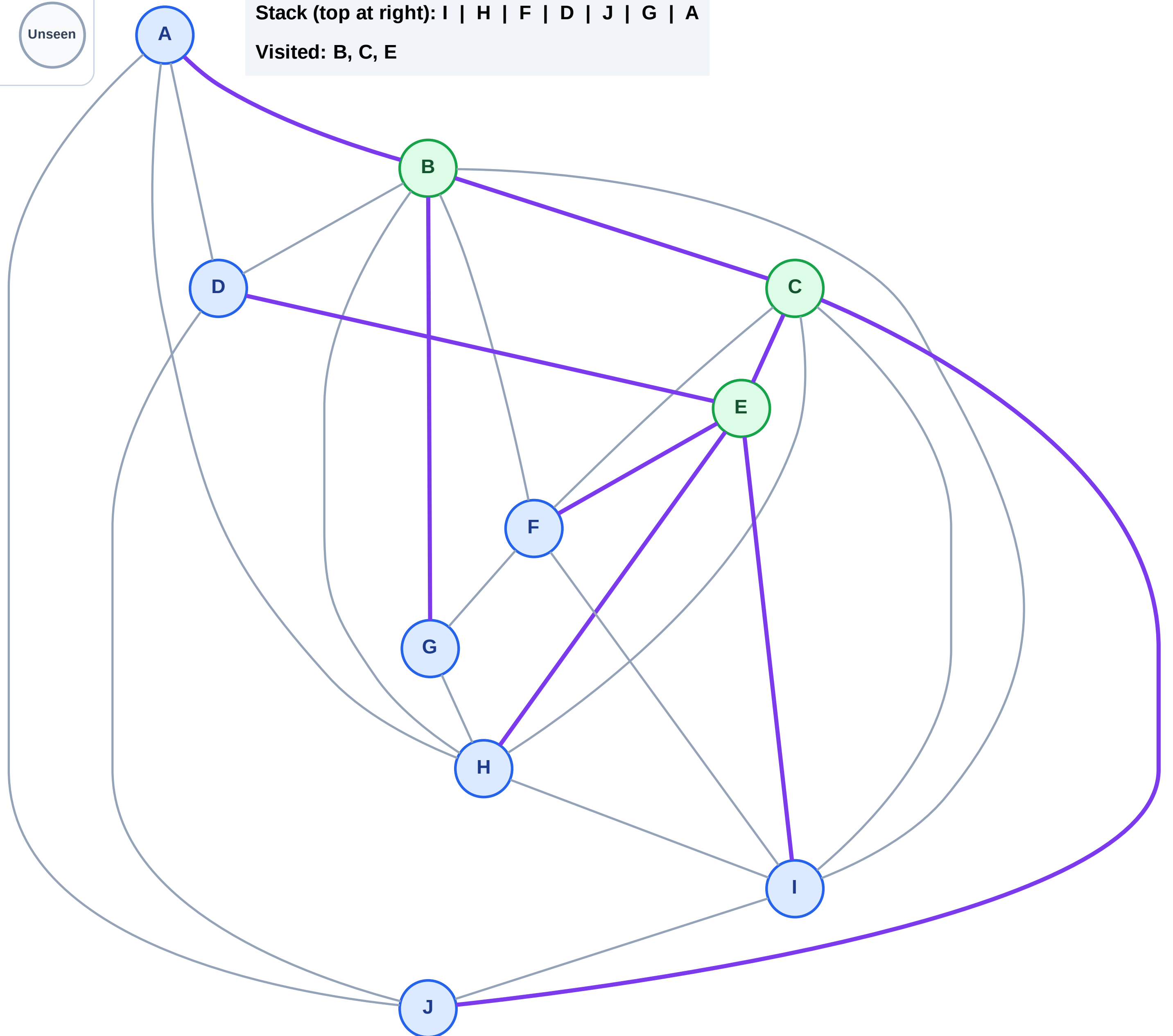
Step 7: Discover G, A from B

Legend



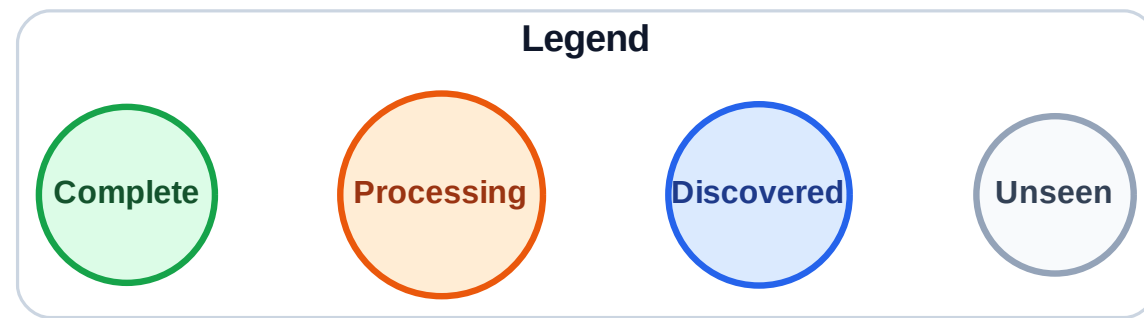
Stack (top at right): I | H | F | D | J | G | A

Visited: B, C, E

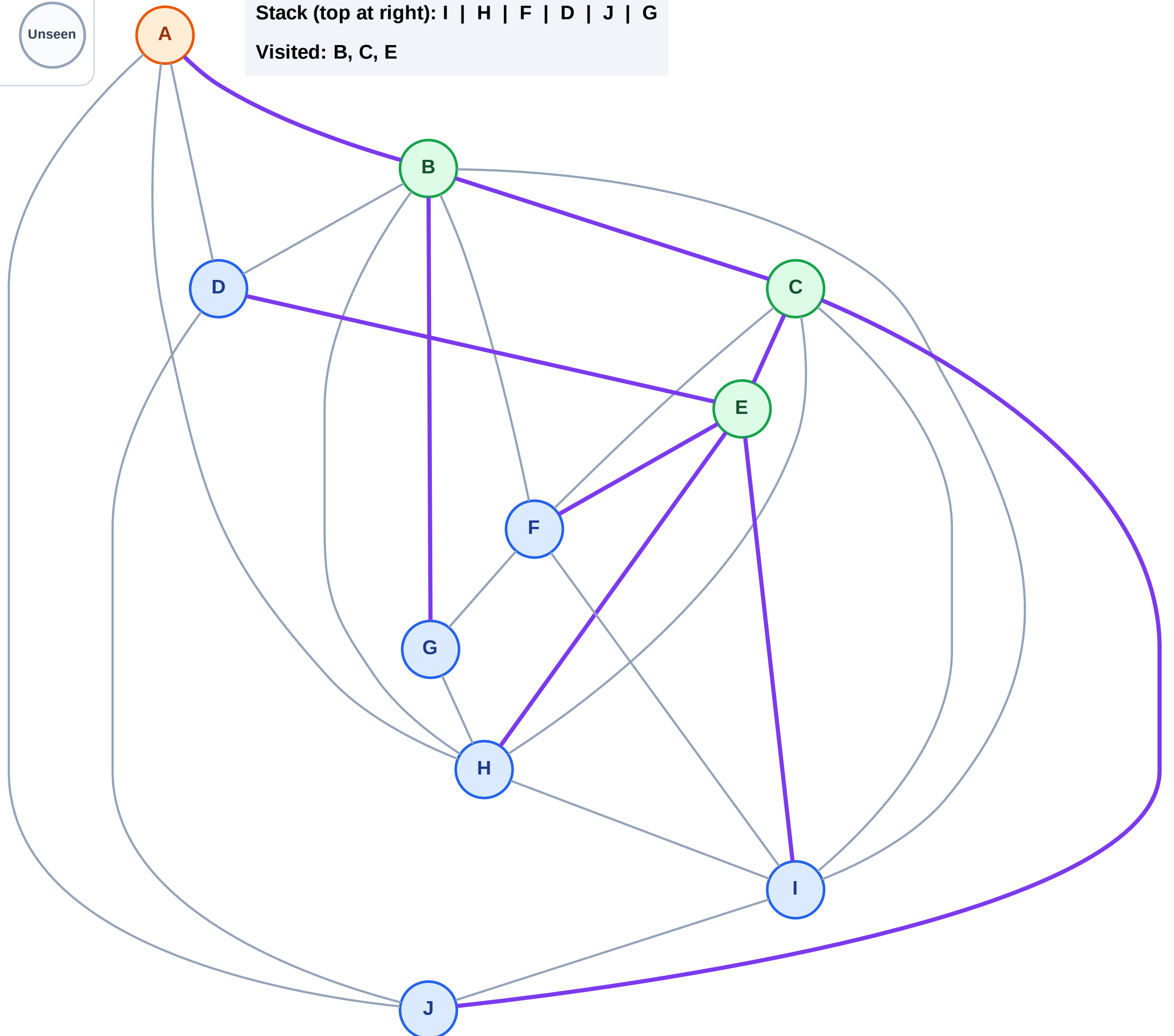


DFS Traversal

Step 8: Process node A

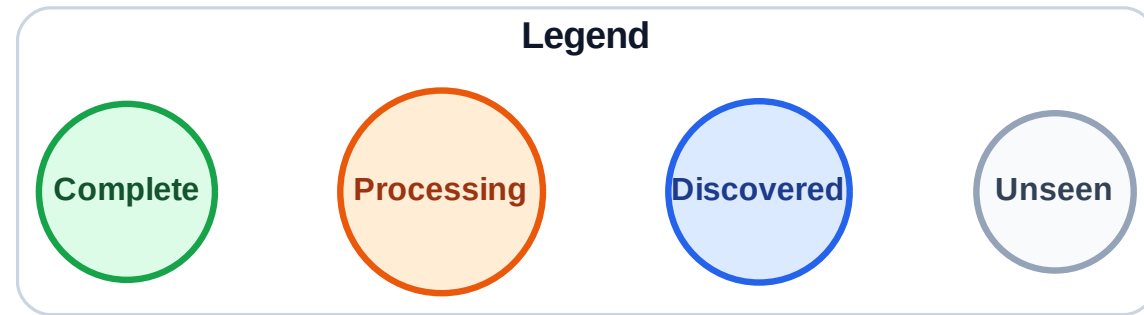


Stack (top at right): I | H | F | D | J | G
Visited: B, C, E

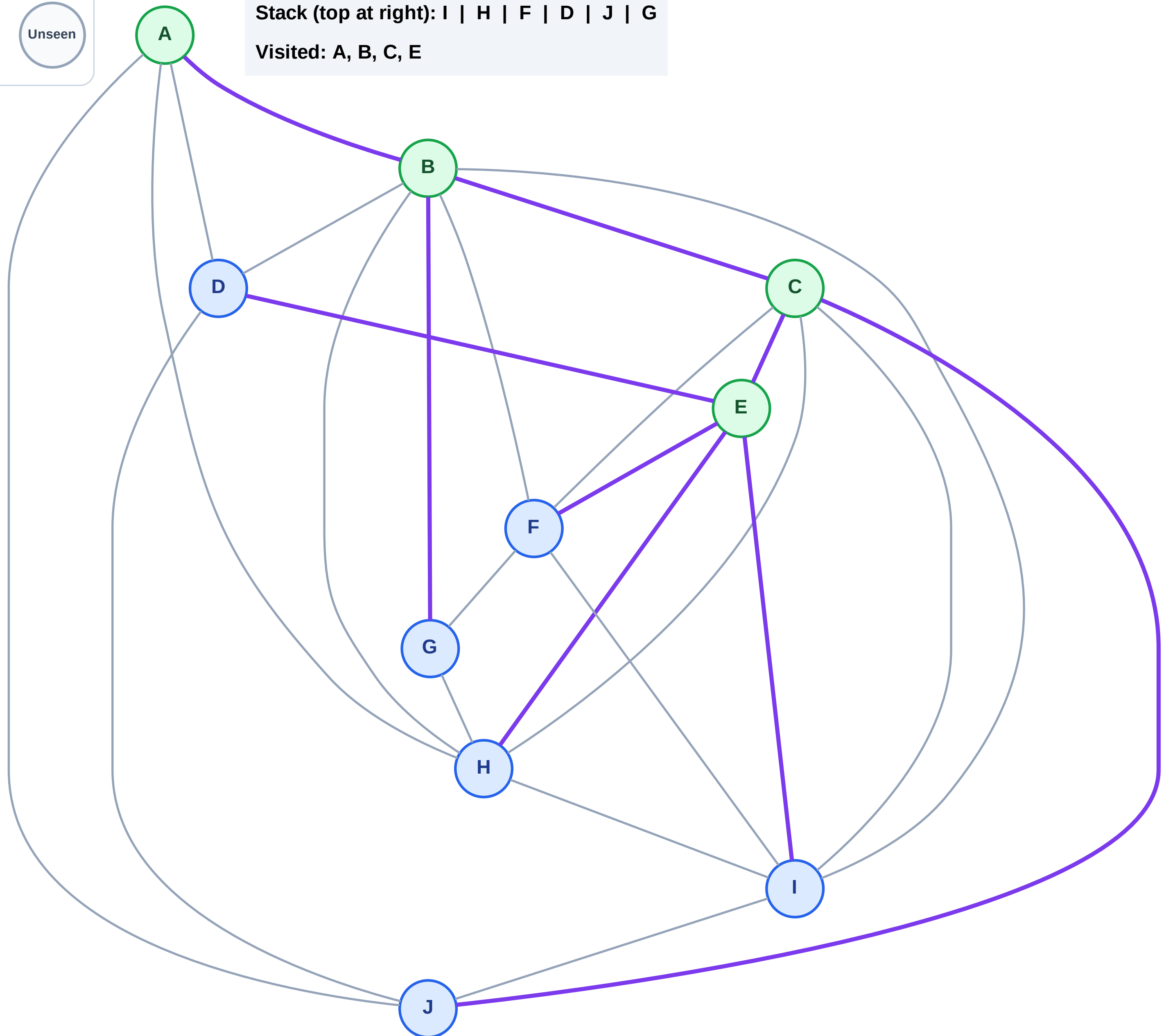


DFS Traversal

Step 9: No new neighbors from A

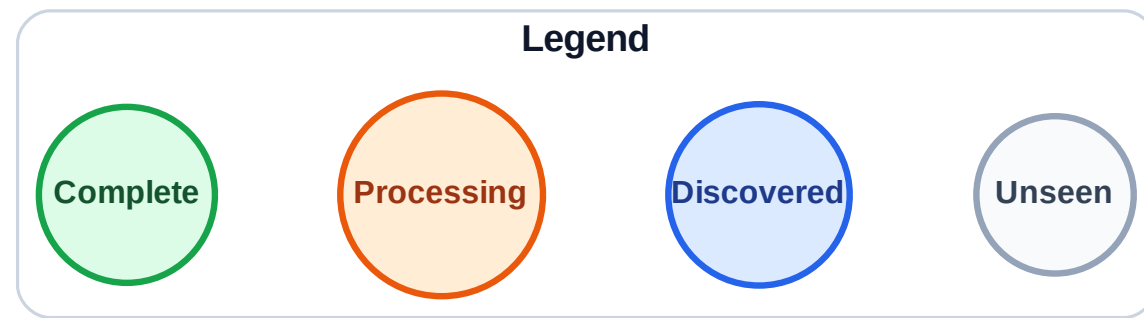


Stack (top at right): I | H | F | D | J | G
Visited: A, B, C, E

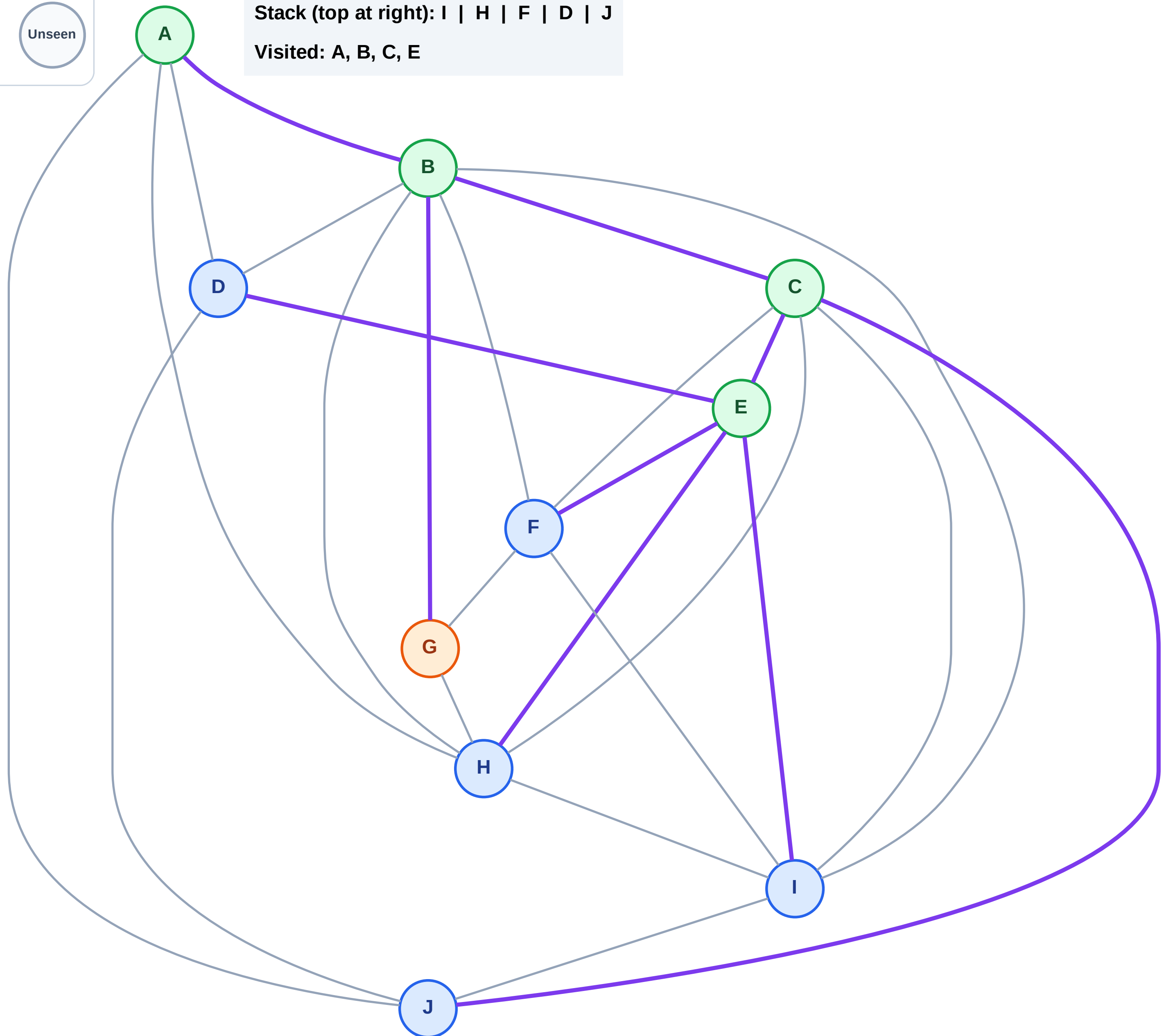


DFS Traversal

Step 10: Process node G

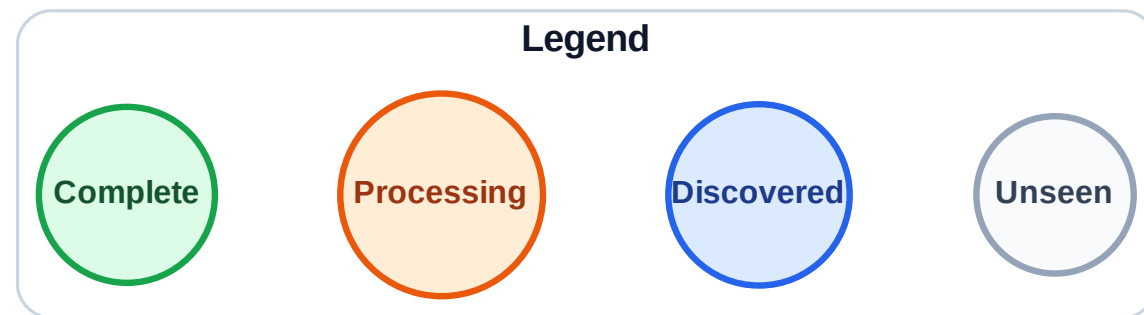


Stack (top at right): I | H | F | D | J
Visited: A, B, C, E

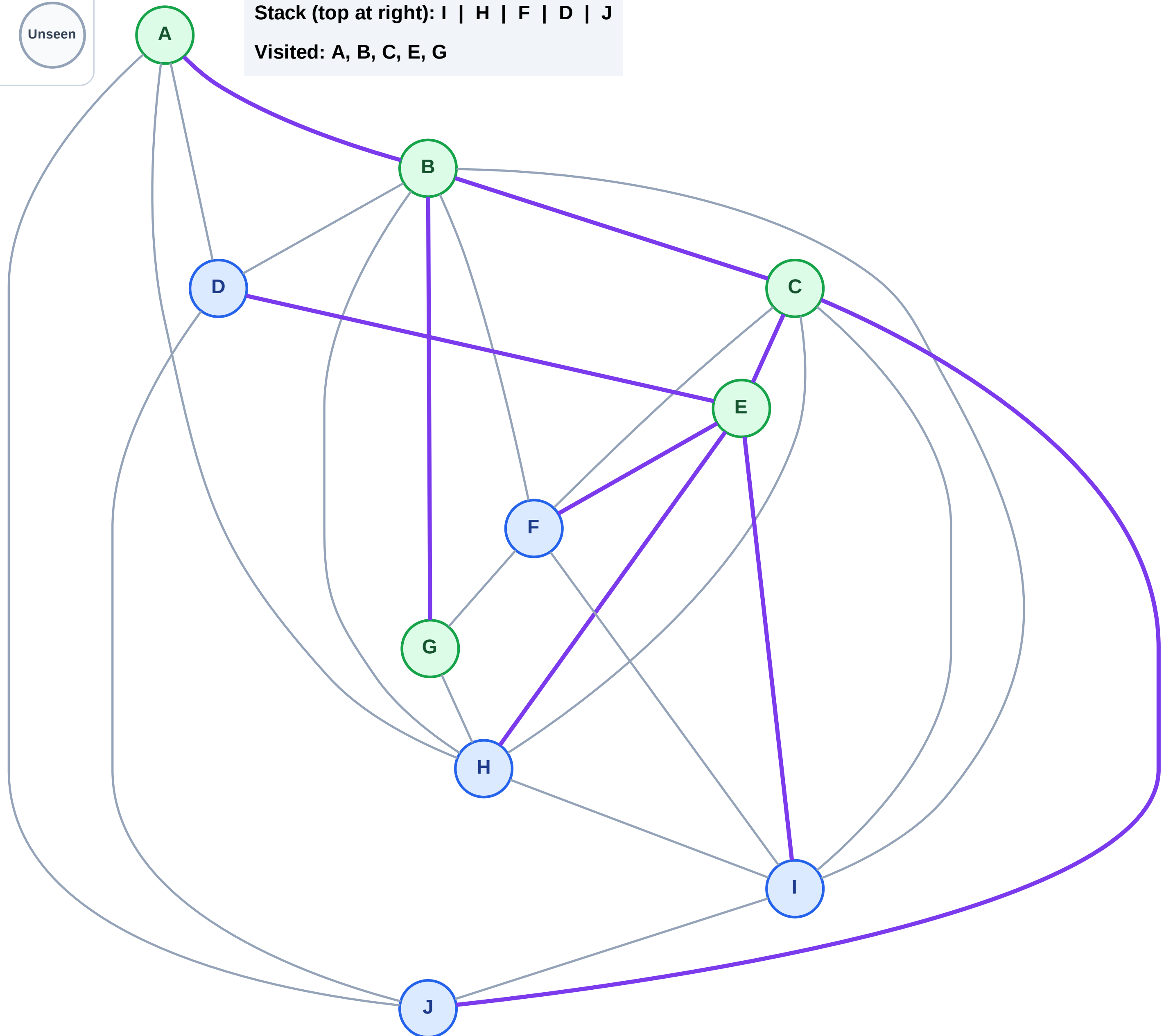


DFS Traversal

Step 11: No new neighbors from G



Stack (top at right): I | H | F | D | J
Visited: A, B, C, E, G



DFS Traversal

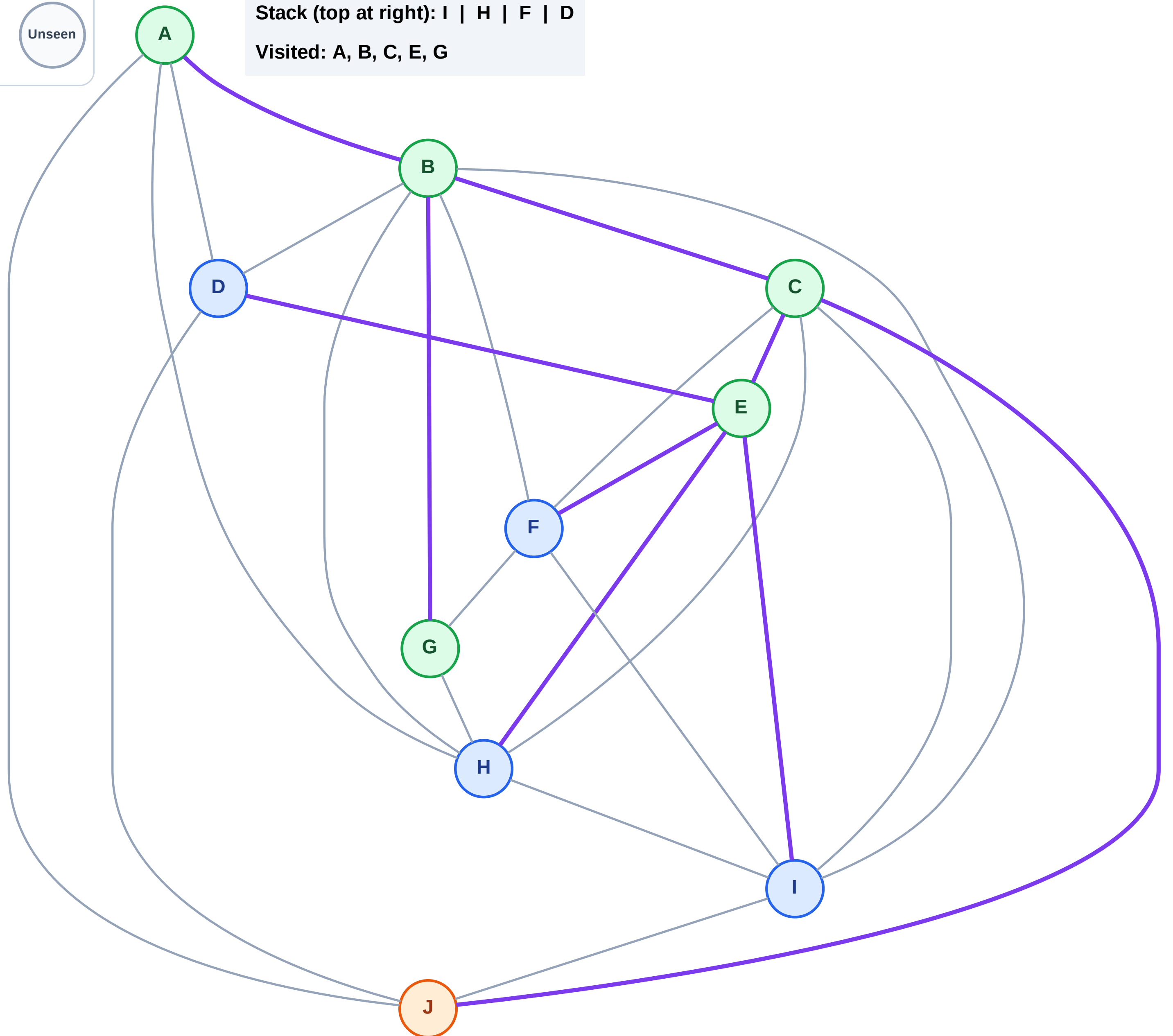
Step 12: Process node J

Legend



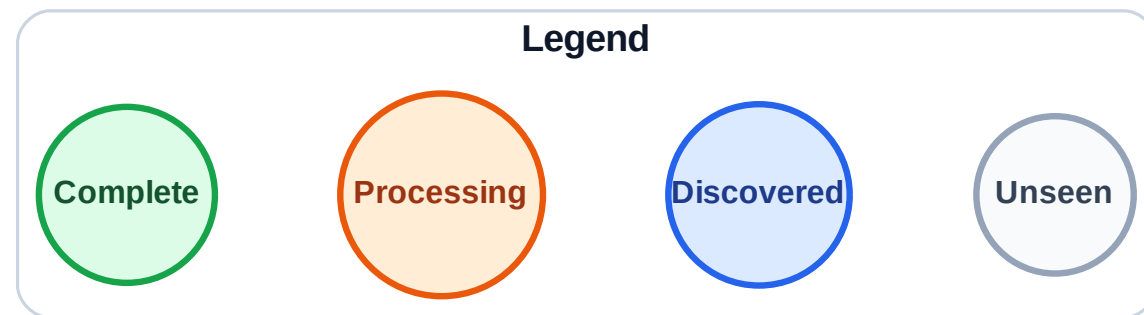
Stack (top at right): I | H | F | D

Visited: A, B, C, E, G

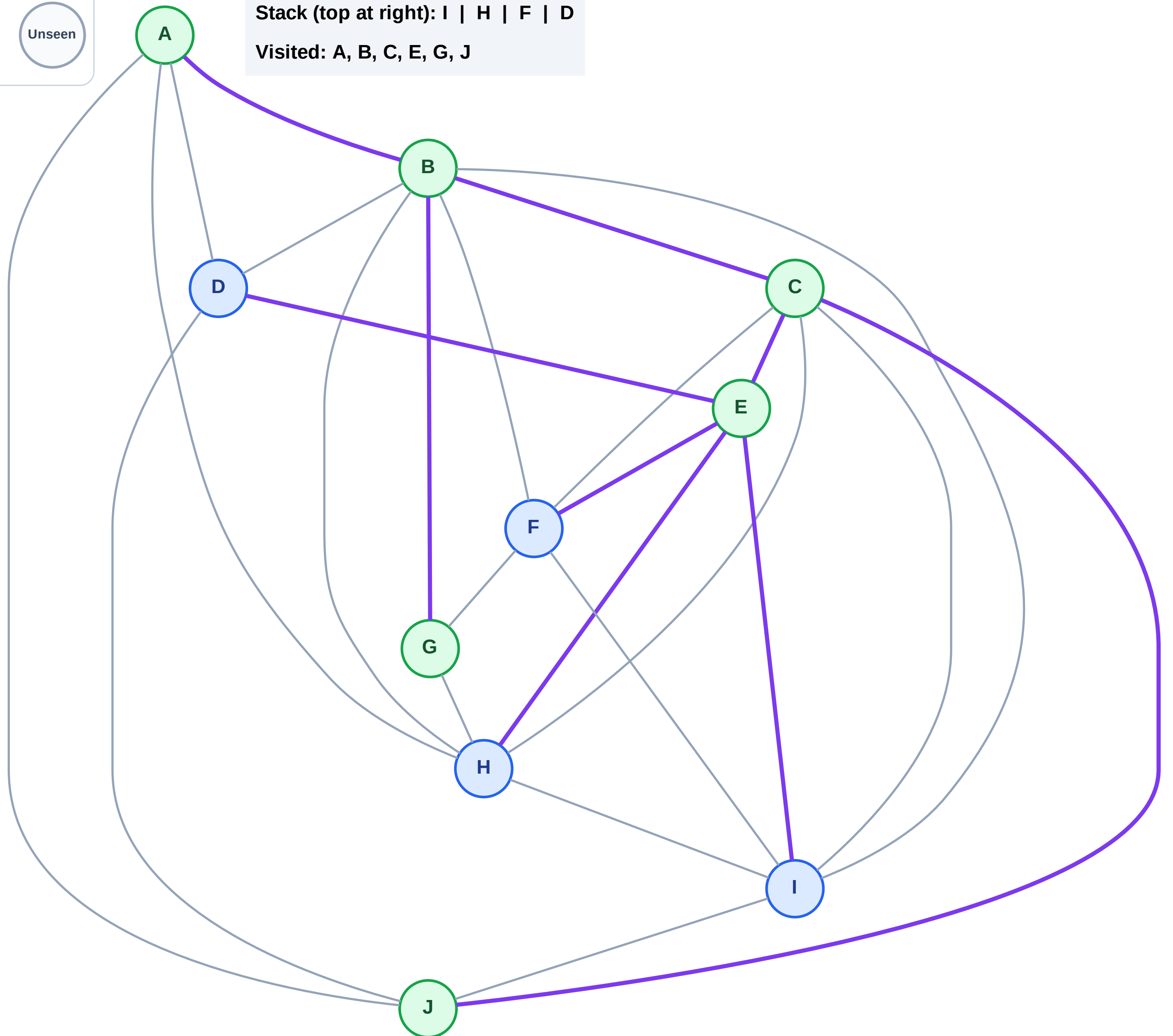


DFS Traversal

Step 13: No new neighbors from J

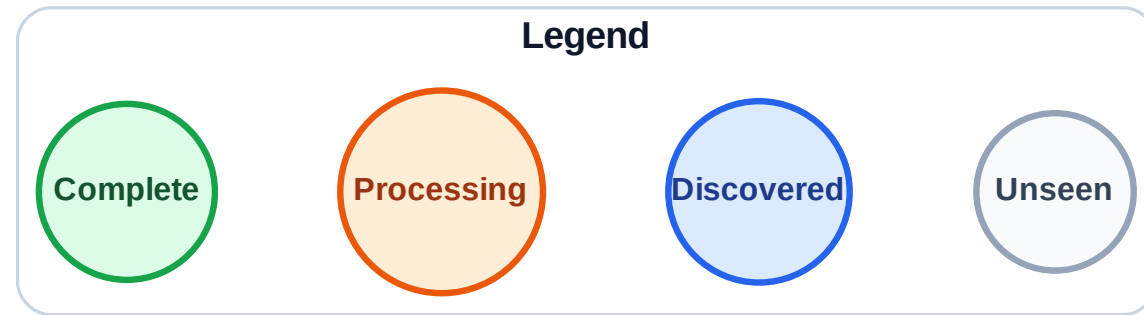


Stack (top at right): I | H | F | D
Visited: A, B, C, E, G, J

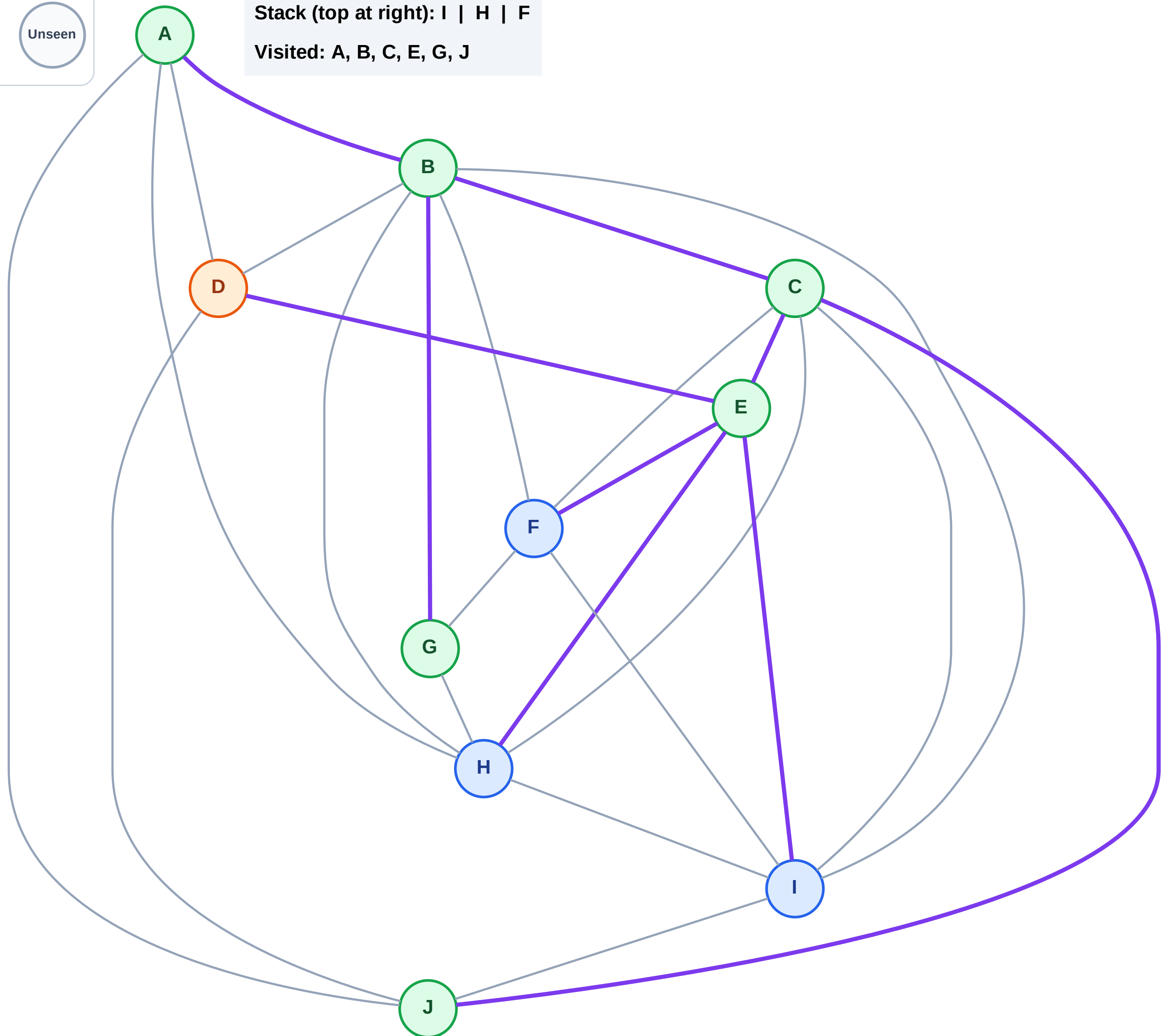


DFS Traversal

Step 14: Process node D

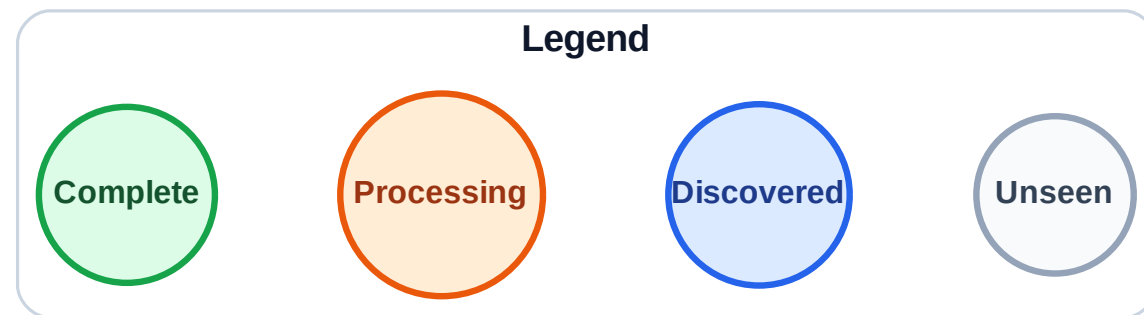


Stack (top at right): I | H | F
Visited: A, B, C, E, G, J



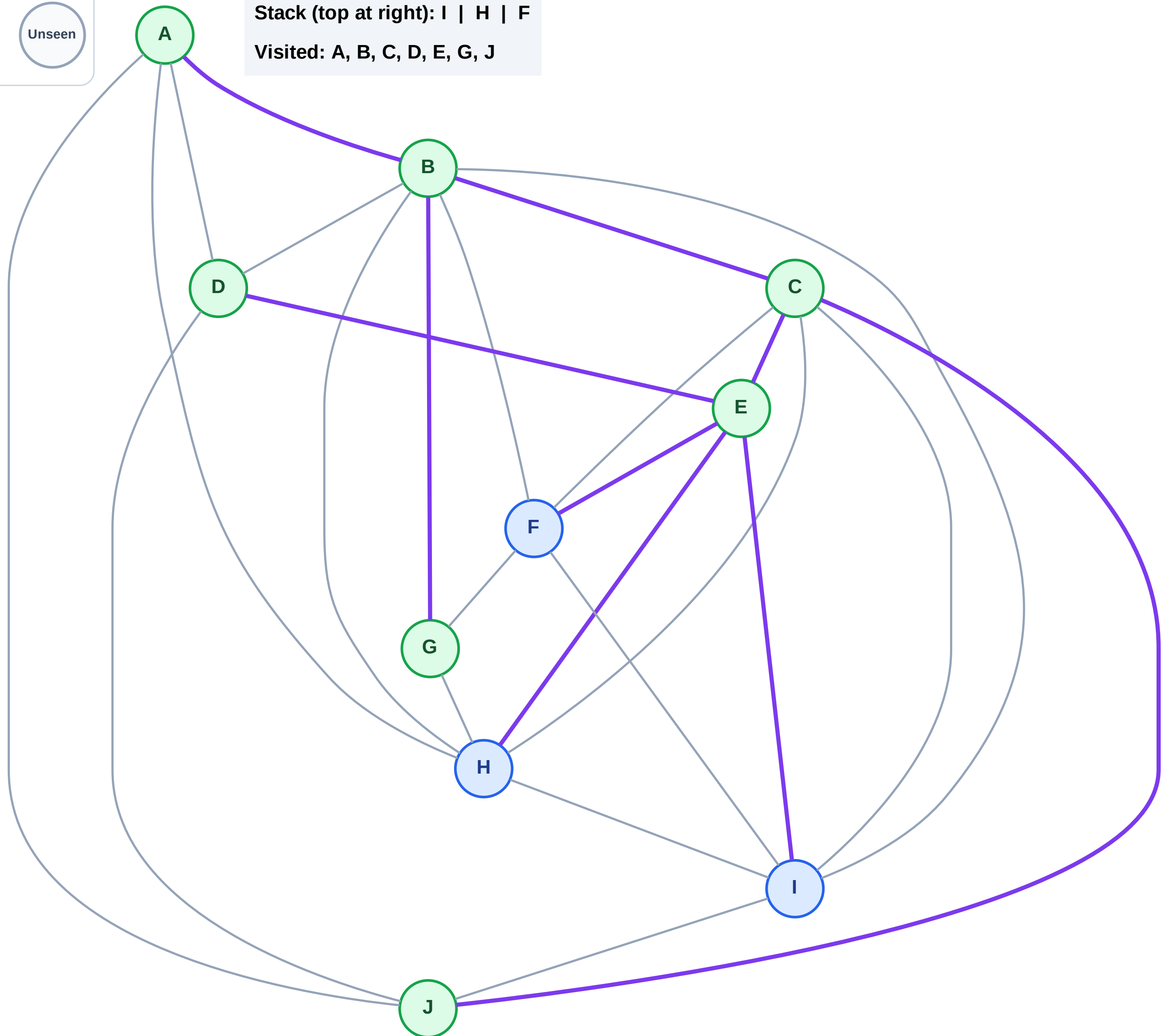
DFS Traversal

Step 15: No new neighbors from D



Stack (top at right): I | H | F

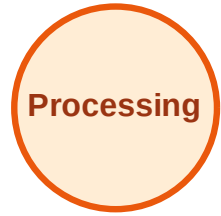
Visited: A, B, C, D, E, G, J



DFS Traversal

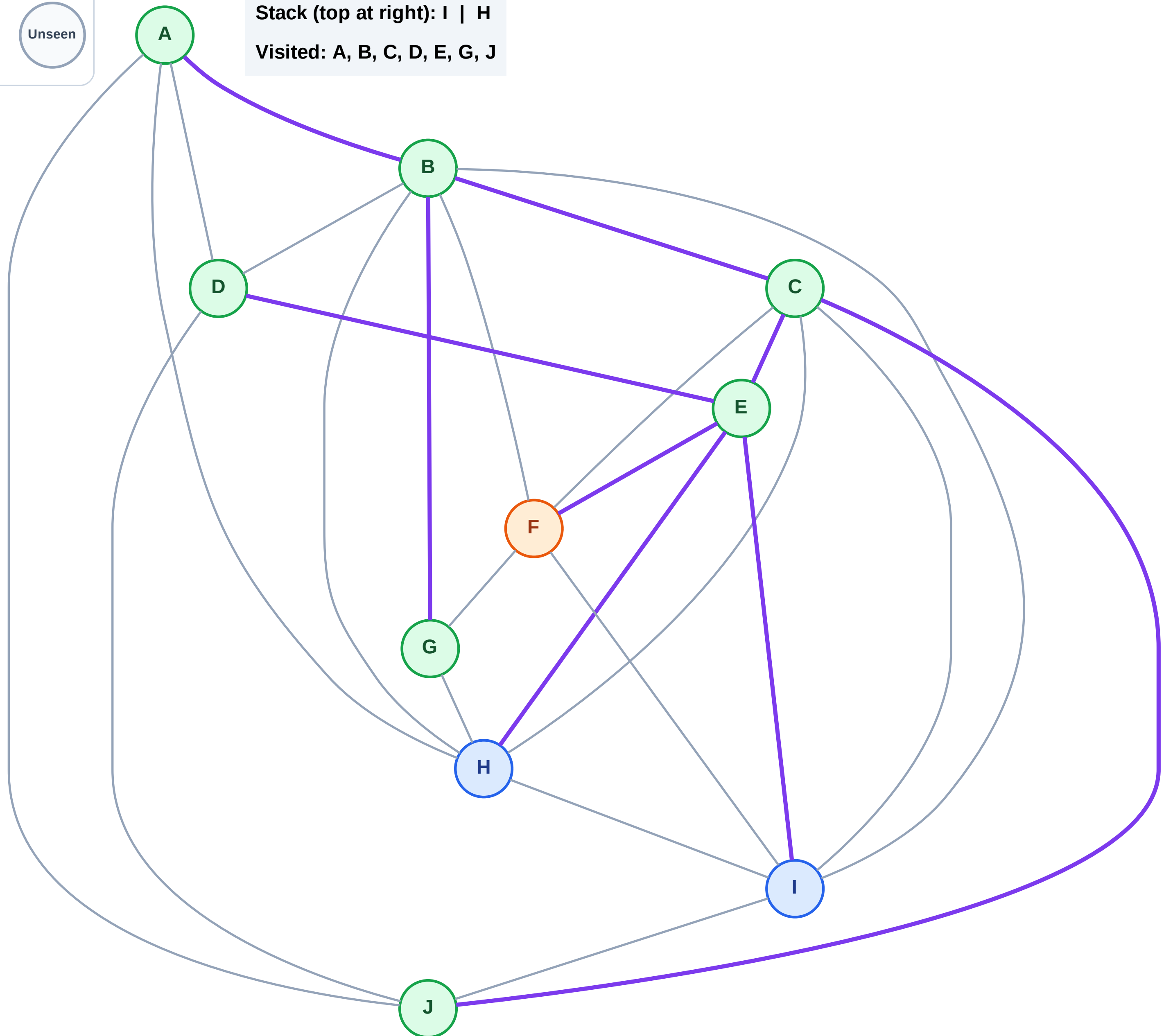
Step 16: Process node F

Legend



Stack (top at right): I | H

Visited: A, B, C, D, E, G, J



DFS Traversal

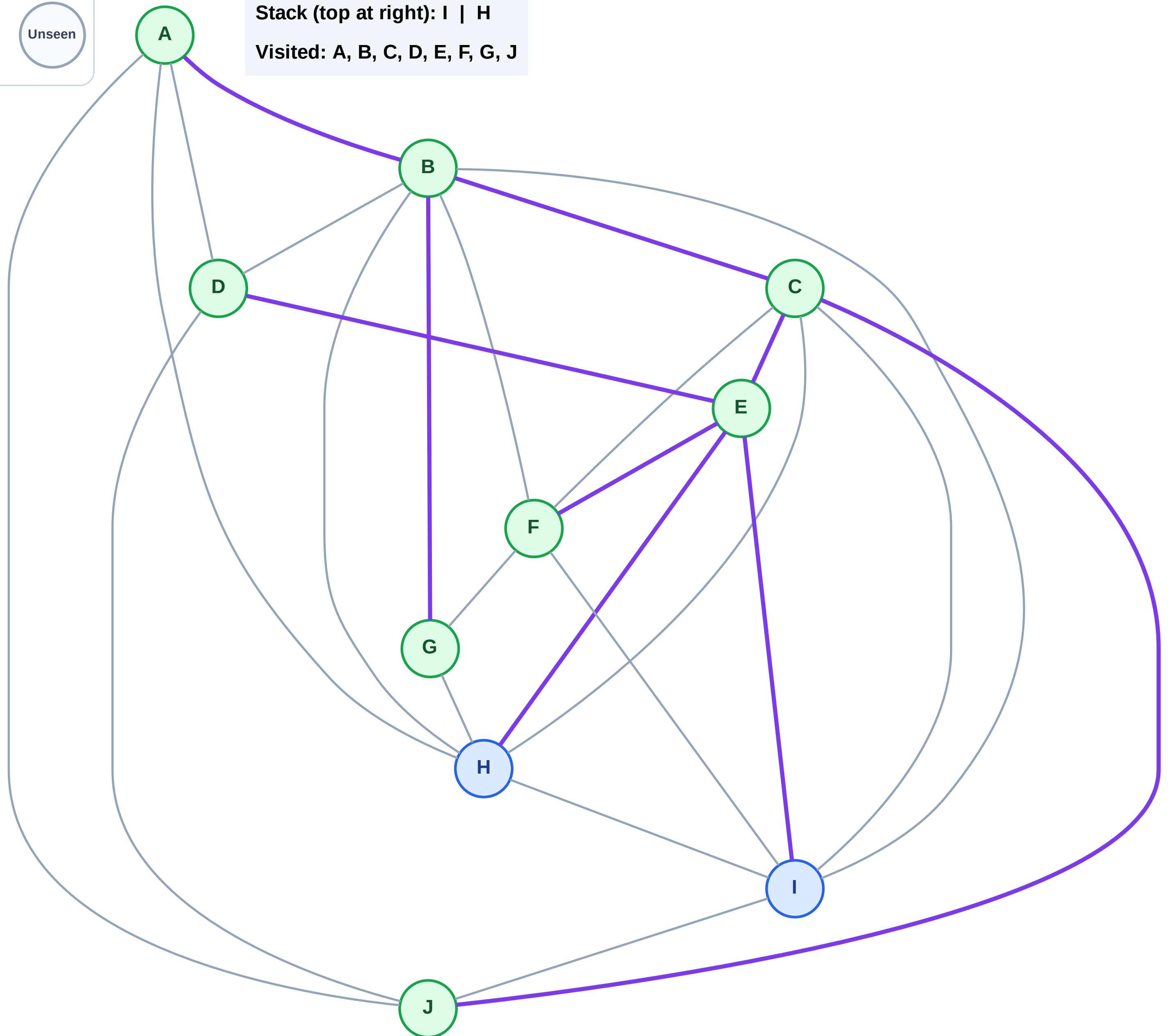
Step 17: No new neighbors from F

Legend



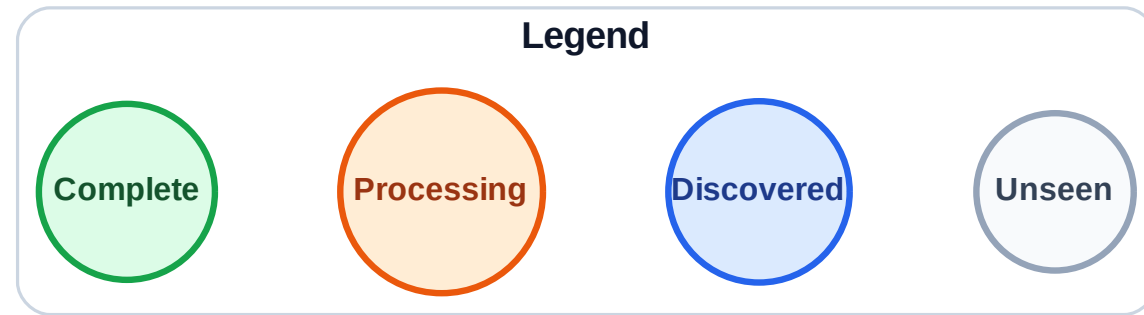
Stack (top at right): I | H

Visited: A, B, C, D, E, F, G, J



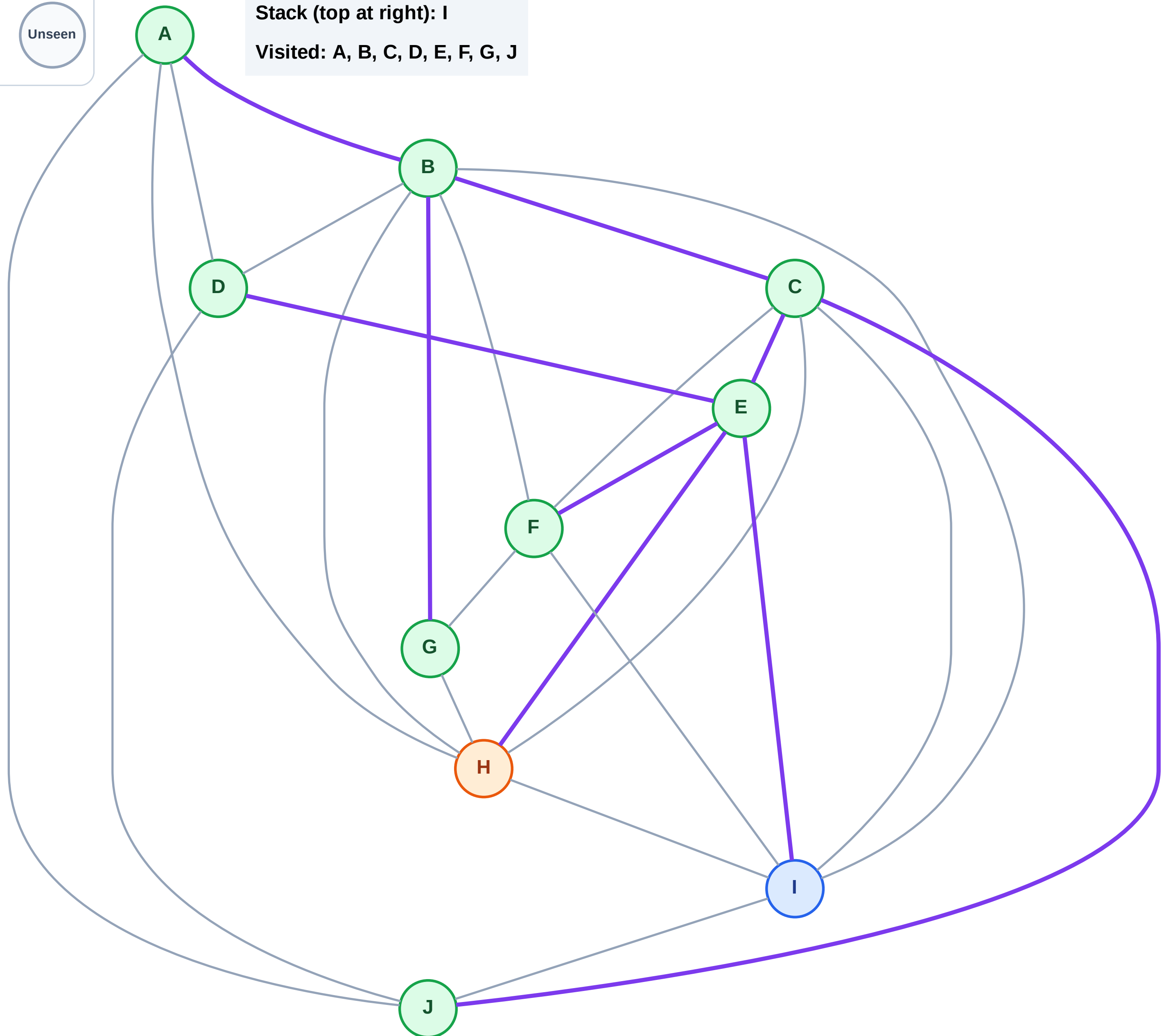
DFS Traversal

Step 18: Process node H



Stack (top at right): I

Visited: A, B, C, D, E, F, G, J

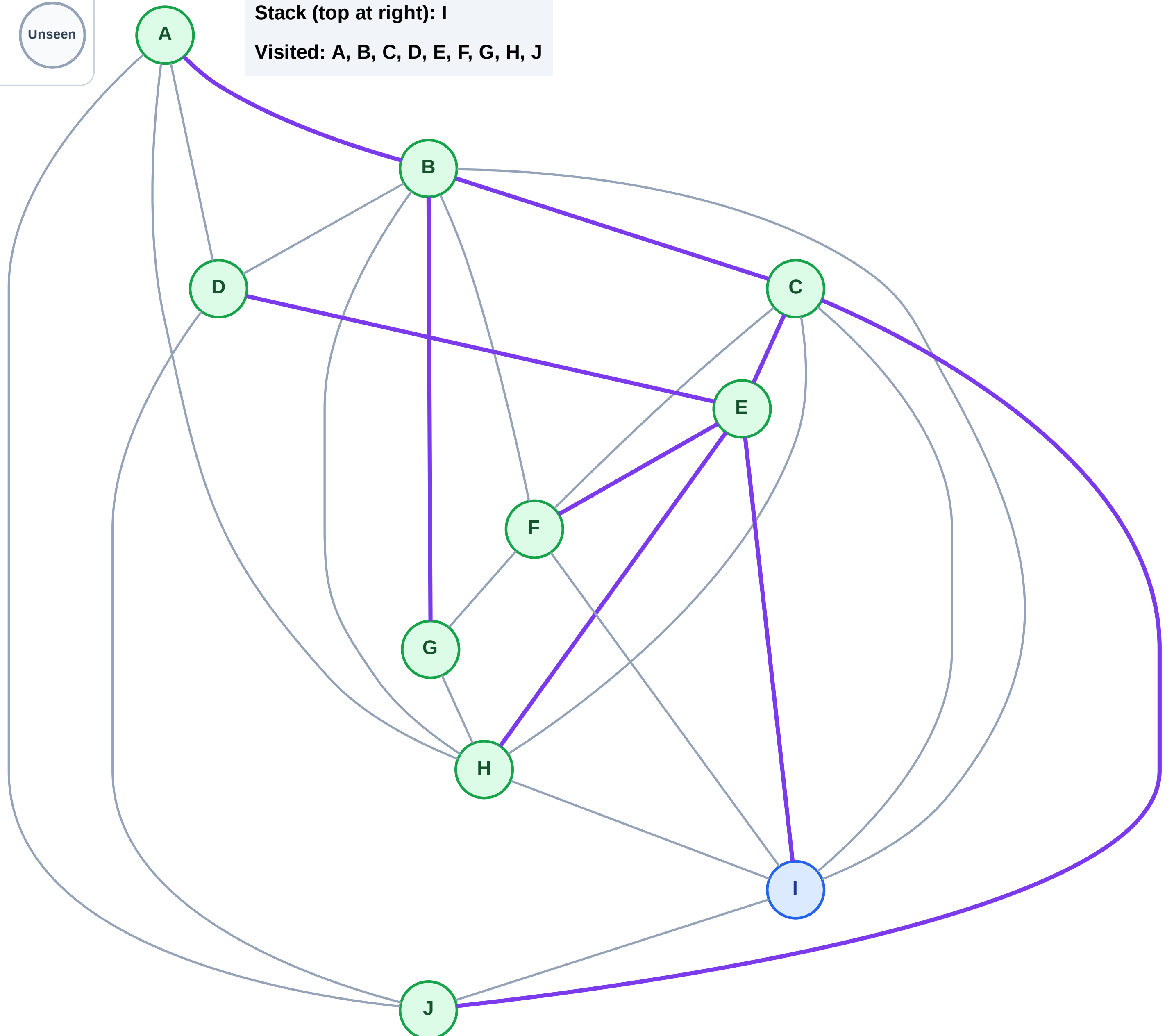
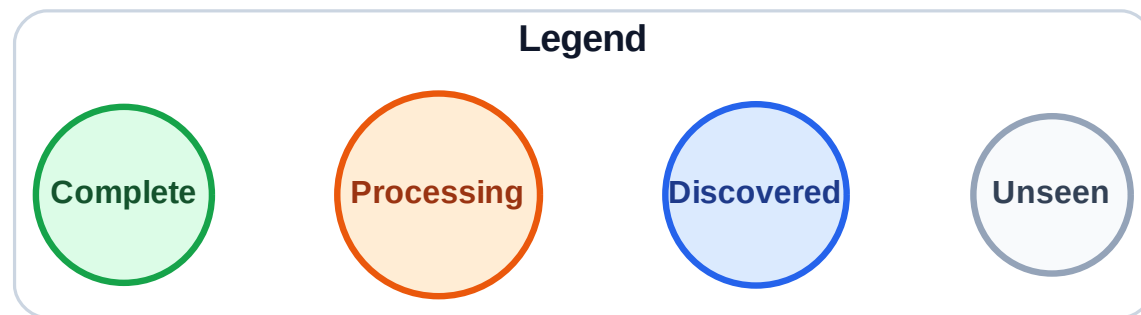


DFS Traversal

Step 19: No new neighbors from H

Stack (top at right): I

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

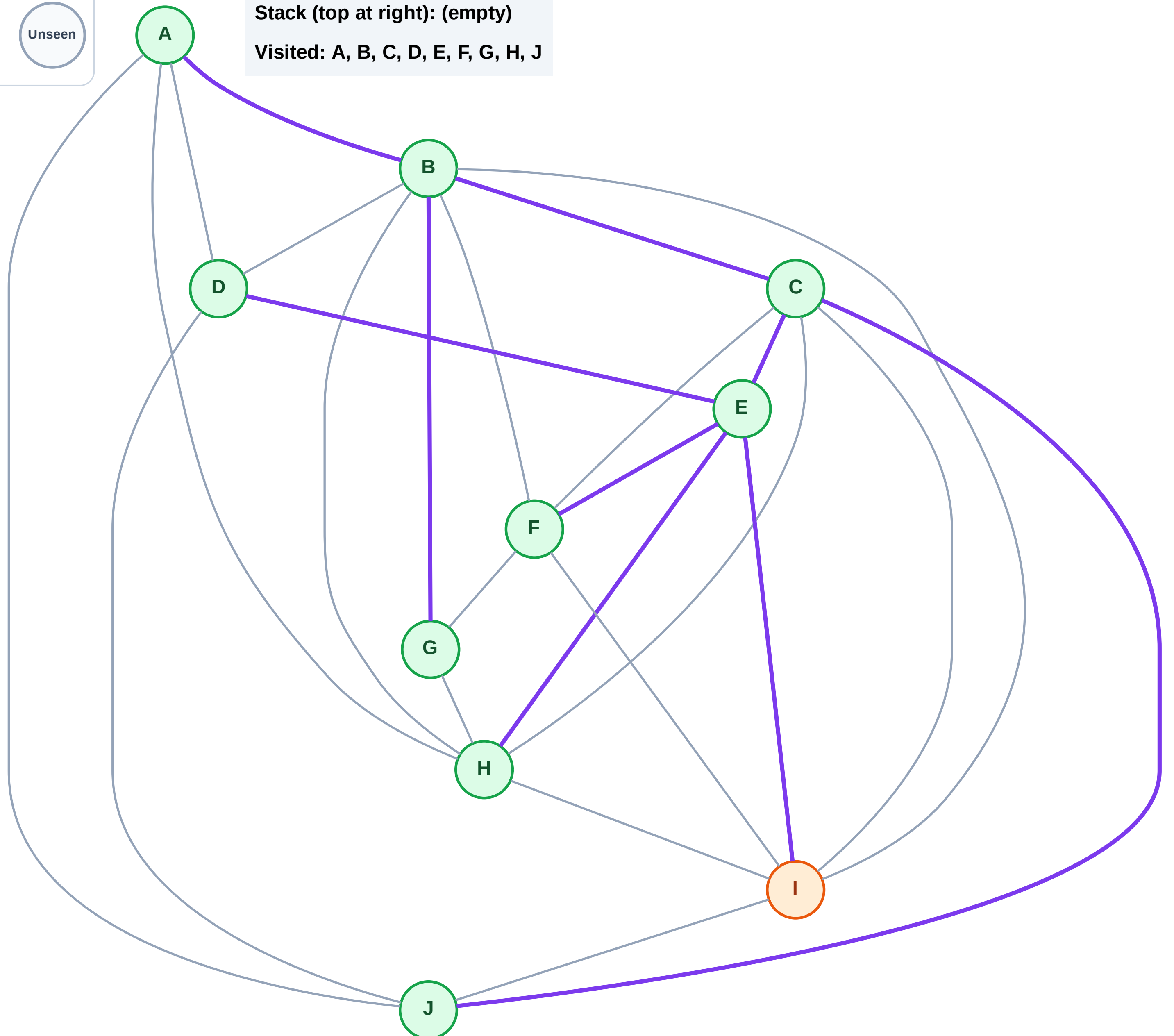
Step 20: Process node I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

